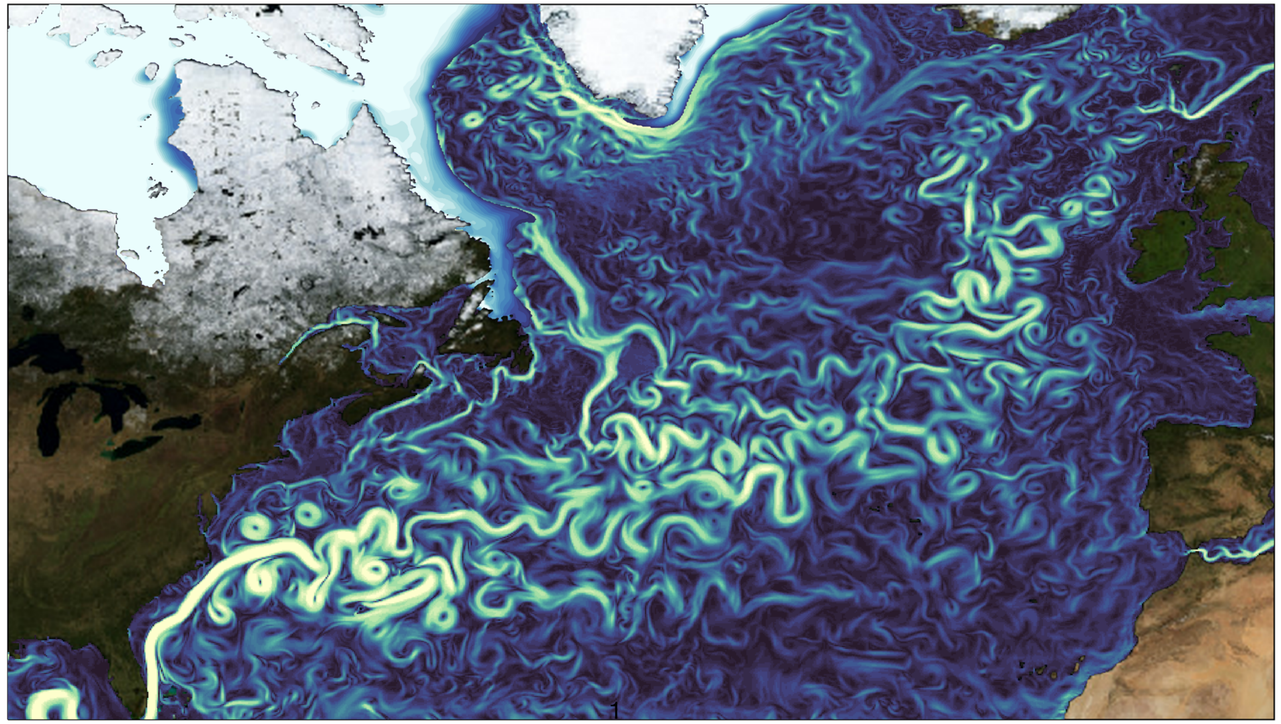
Problems and prospects at the frontier of ocean climate model theory and practice

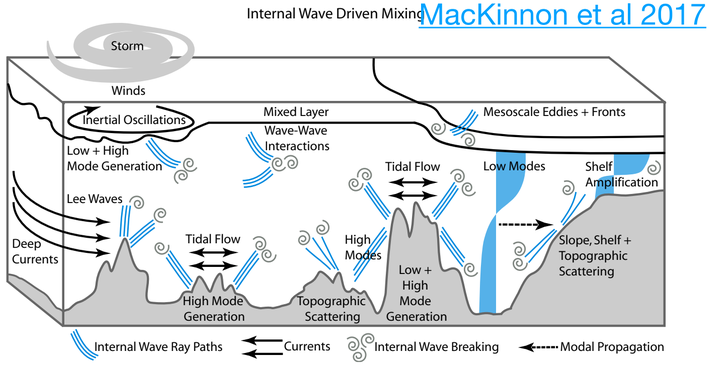
Stephen Griffies
IPSL/LOCEAN + Princeton University
Virtual talk given to the STUOD group at Imperial College
18 Sept 2026



Some goals for ocean climate model developers

Numerical methods that respect properties of the continuum.

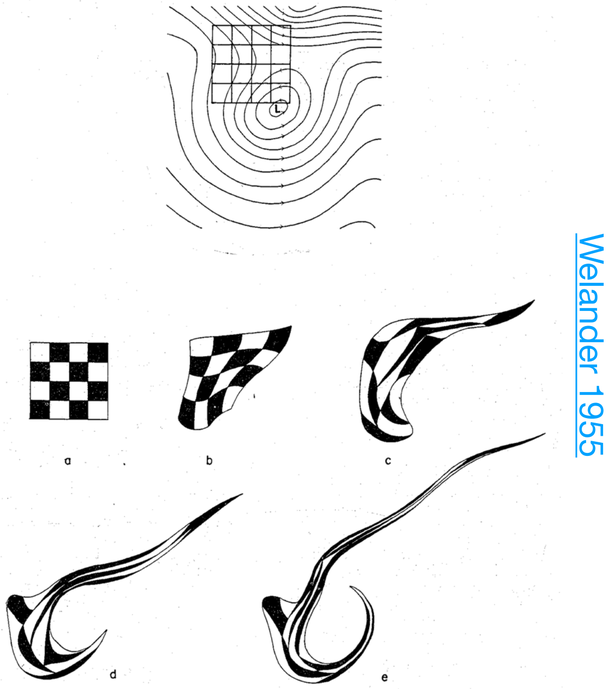
- Global and local conservation of scalars: mass, enthalpy, salt, carbon, nutrients
- Momentum/Force consistency: including boundaries/topography
- Energetic consistency: conversions between energy reservoirs
- Vorticity consistency: including the representation of boundary torques
- Nonlinear cascades: direct at small (3d) scales; inverse at large (QG) mesoscales



MacKinnon et al 2017

Methods that respect physical levels of interior ocean mixing

- Outside boundary layers, the ratio of neutral mixing (along buoyancy surfaces) to dianeutral (across buoyancy) mixing is roughly 10^8: a huge scale separation!
- This tiny level of mixing must be respected even in the presence of unstable flows that lead to active geostrophic turbulence, with the associated downscale cascade of tracer variance.
- For the climate problem, we must respect physical mixing levels for 1000s of eddy turnover times! This is an incredibly tough computational physics problem!



Welander 1955

Low Carbon footprint and efficient turnaround on science

- Try not to contribute to the global warming we are studying!
- Nor can we wait too long to run a simulation worth analyzing!

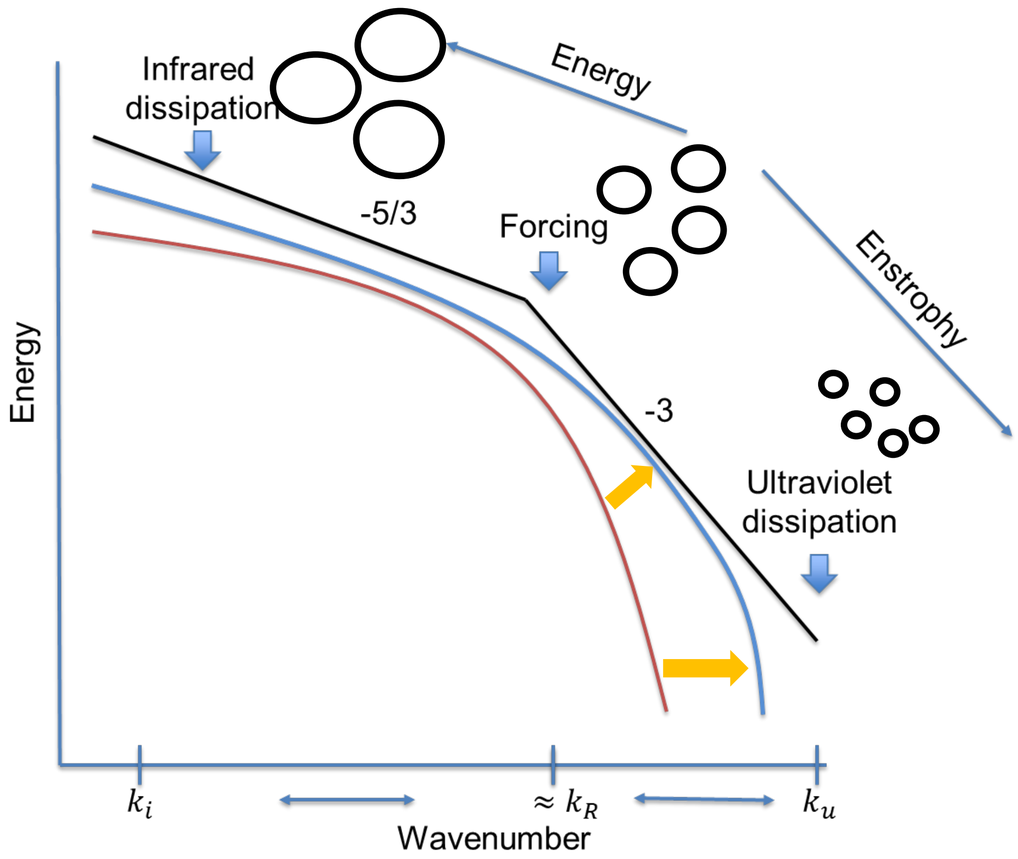
Take-home themes/messages

- **Kinetic energy backscatter methods**
  - Methods that re-inject kinetic energy that otherwise disappears from the simulation through eddy viscosity.
  - They are relevant to eddy-active models whose inverse cascade is truncated due to grid limitations.
  - Backscatter is not a direct physical parameterization; instead, it is a reorganization of energy pathways.
  - Unresolved questions/problems remain, and I conjecture they relate to flow/boundary representation.
- **Spurious diapycnal mixing + representation of vertical (upward) eddy heat transport**
  - Spurious mixing happens, in varying quantities, within all ocean climate models.*
  - Weak vertical eddy heat transport happens when under parameterize and/or under represent mesoscale.
  - Each issue contributes to problems with ocean heat uptake, long-term model drift, global sea level, etc.
  - Mitigation requires smart vertical coordinates, eddy parameterizations/representations, and accurate numerical transport.
  - Ongoing issues with locally measuring spurious mixing; constructing vertical coordinates; enhancing numerical accuracy in advective transport operators; improving eddy parameterizations/representations…
- **Vorticity-aware methods**
  - Vorticity is a diagnostic field. It is not a prognostic field. It is central to how we think about ocean processes, including its waves, instabilities, turbulence, and circulation.
  - Smart methods can improve the numerical representation of vorticity, but there is a lot of work needed.
  - Conjecture: further enhancements to vorticity-aware methods will improve boundary current representation, given that such currents interact strongly with the solid boundary and these boundaries are key to forcing ocean vorticity.

*Stacked shallow water models have no spurious diapycnal mixing. But these are not climate models. They are idealized process models.

Part I: Kinetic energy backscatter

Basics of kinetic energy backscatter



[2022 talk from Stephan Juricke](#) (Kiel)

$$\partial_t \mathbf{u} + (f + \zeta)\mathbf{k} \times \mathbf{u} + \nabla(K + M) = \mathbf{F}_v + \mathbf{F}_h,$$

$$\mathbf{F}_h = -\nabla \cdot \left[\nu_4 \nabla\left(\nabla^2 \mathbf{u}\right)\right] + \nabla \cdot (\nu_2 \nabla \mathbf{u}).$$

The harmonic viscosity ν_2, which is negative to represent backscatter

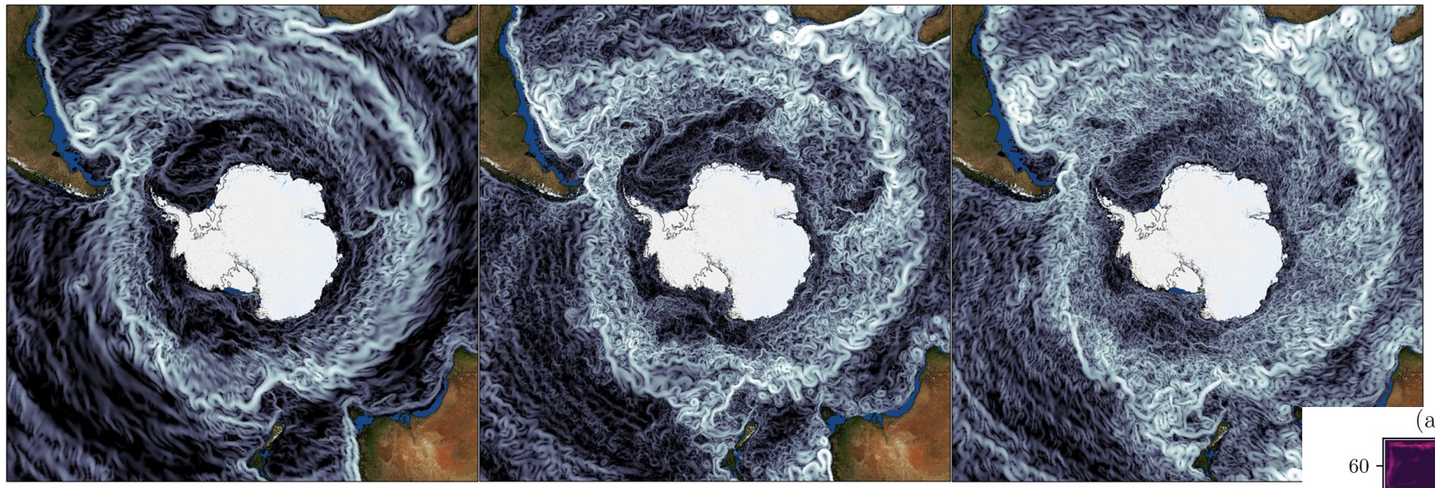
``The combination of a negative Laplacian viscosity with a biharmonic viscous dissipation may be viewed as the simplest expansion of the sub-grid eddy effects in terms of isotropic (hyper-)diffusivities, which can fulfill the constraints of enstrophy dissipation and energy conservation.'' [Jansen and Held (2014)](#)

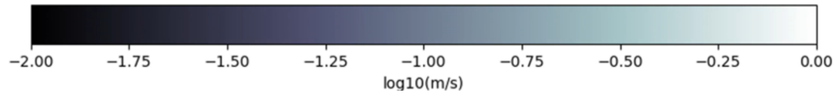
``Backscatter reduces overdissipation by reinjecting a portion of the excessively dissipated kinetic energy into the resolved flow. Reinjecting this energy on larger scales feeds the inverse energy cascade which is partly resolved in eddy-permitting simulations... As the physical rate of enstrophy dissipation is maintained, backscatter proves to be numerically stable despite its antidiffusive behavior on intermediate scales'' [Juricke et al 2019](#)

Example effects from backscatter

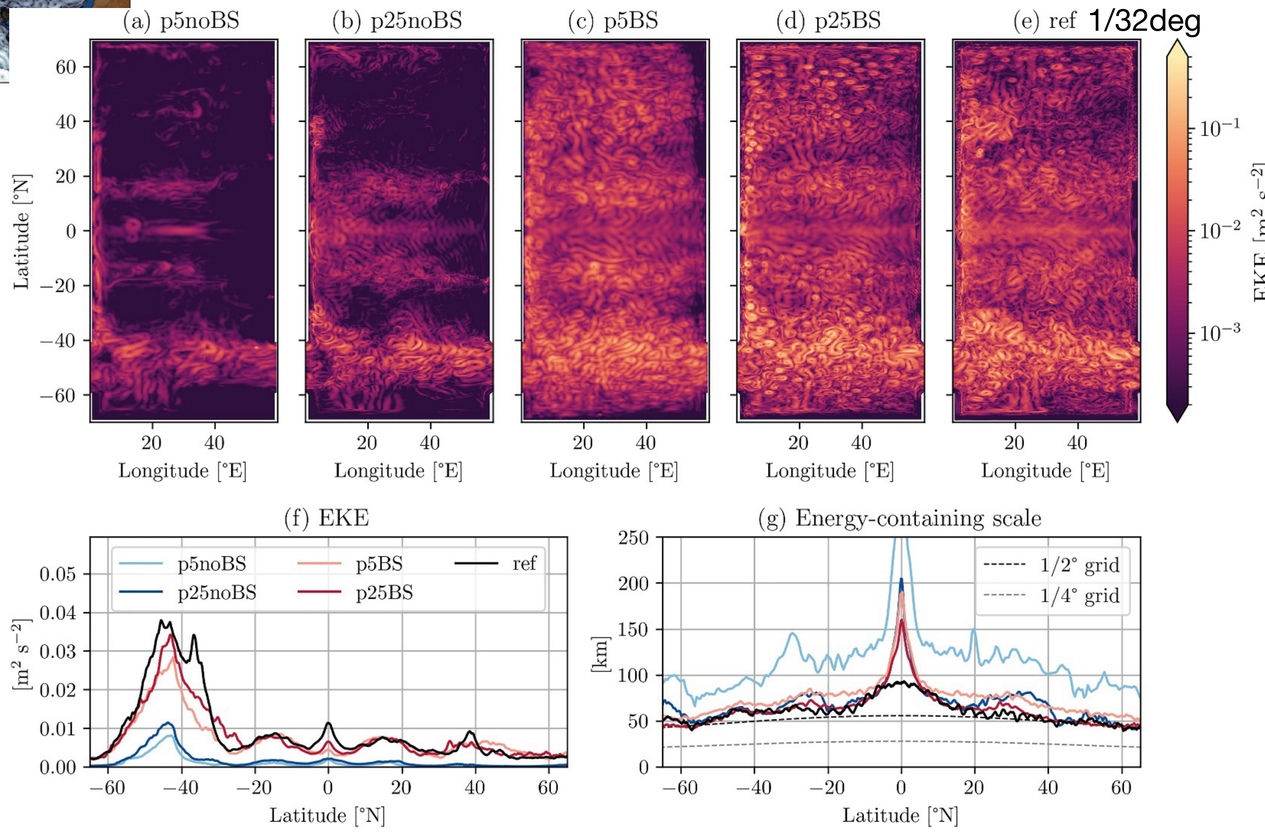
Juricke talk (2022)

25km | Eddy resolving | 25km with backscatter





More energetic flows + finer scale flow features



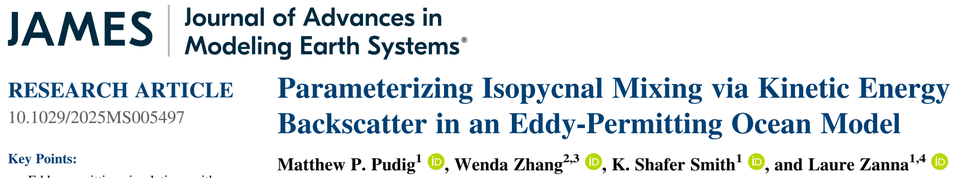
Pudig et al (2026)

Further intriguing advances seen in eddy-permitting models

Yankovsky et al (2024): Backscatter removes the need for a Gent-McWilliams-type operator at the eddy-permitting scales (25 km or finer).	``(a) both the potential and kinetic energy effects of eddies may be parameterized via a kinetic energy backscatter, with no Gent-McWilliams along-isopycnal transport; (b) a dominant factor in ensuring a physically-accurate backscatter is the vertical structure of the parameterized momentum fluxes.”
Pudig et al (2026): Backscatter removes the need for a neutral diffusion (Redi) operator at the eddy-permitting scales (25 km or finer).	``Our results suggest that parameterizing backscatter can plausibly capture key mesoscale physics in a unified framework: the inverse cascade of kinetic energy, the slumping of steep isopycnals, and the along-isopycnal mixing of tracers.”

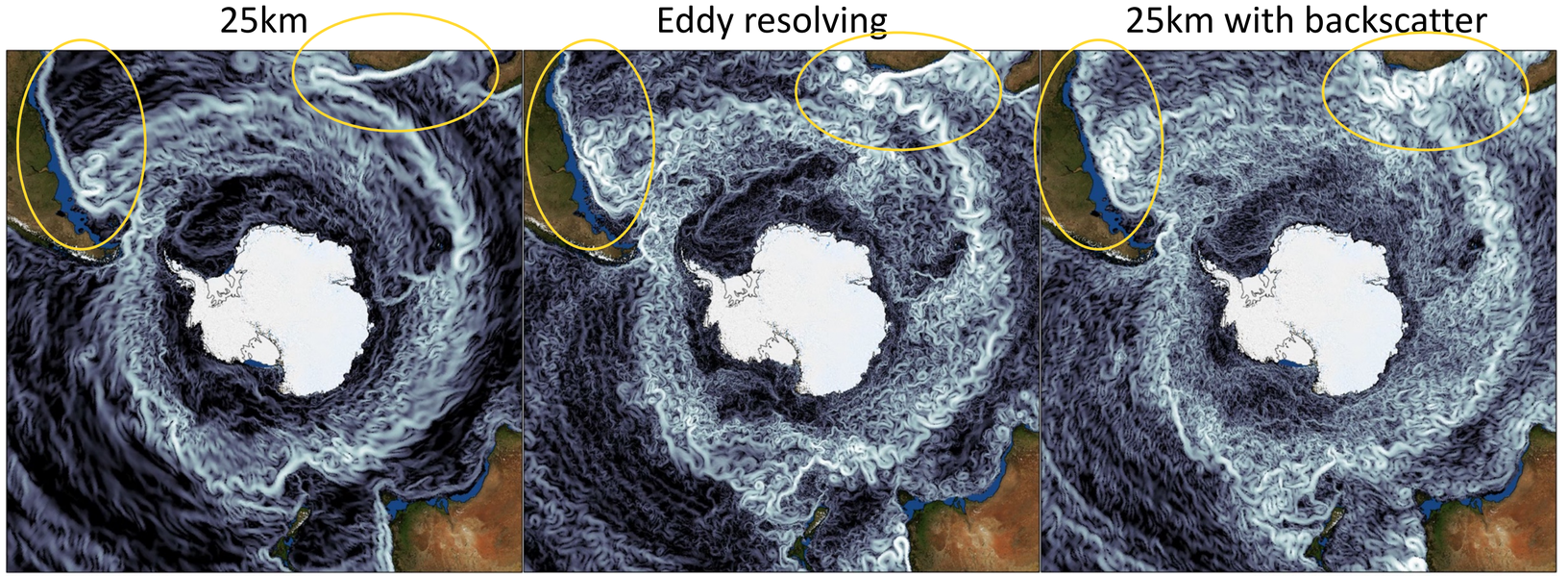
These results are based on idealized simulations with stacked shallow water (MOM6 Neverworld2 configuration).

They need to be pursued in global climate-scale simulations. If robust, it greatly reduces the cost, formulation/ tuning, and analysis of eddy-permitting simulations.

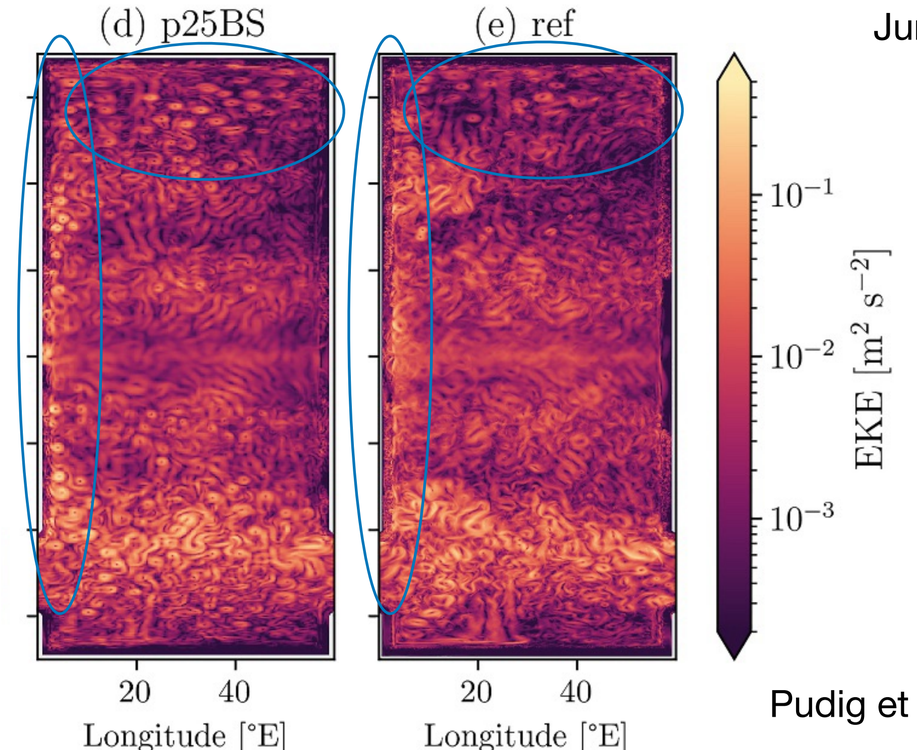


But…there remain some outstanding questions to resolve

- ✔ More energetic flows + finer-scale flow features.
- ✖ But there is an increased and unrealistic build-up of coherent eddy features near western boundaries and in some open ocean regions.
- In nature: inverse cascade is altered/halted near boundaries (esp. western) due to shelf friction + boundary waves.
- But the backscatter methods are based on inverse cascades on unbounded domains, and have no topographic information built-in.
- Further basic research is needed to incorporate boundaries/ topography into the backscatter methods.



Juricke talk (2022)



Pudig et al 2026

Kinetic energy backscatter: summary points

- Backscattering ameliorates the spurious effects from kinetic energy dissipation of the mesoscales due to traditional viscous friction.
- Backstatter allows the numerics at the resolved scales to have access to this otherwise lost kinetic energy.
- Backscattering offers a means to enhance a simulation's incomplete representation of the inverse cascade, so it is particularly relevant for eddy-permitting models (models in the ``gray zone").
- Backscatter is not a direct physical parameterization; instead, it is a reorganization of energy pathways.
- There are unanswered questions/problems arising from too much energy pumped into coherent eddy features, notably at the western boundaries (and at some high-latitude regions).
- Conjecture: too many coherent eddy features are related to flow/boundary interactions that are ignored when formulating backscatter. The inverse cascade is not dominant at boundaries.

Part II: spurious diapycnal mixing from numerical advection operators

Basics of spurious mixing from numerical advection

The numerical representation of advection $= \nabla \cdot (\rho\, C\, \boldsymbol{v})$ generally introduces spurious mixing and unmixing due to truncation errors

$$\nabla \cdot (\rho\, C\, \boldsymbol{v})_{\text{model}} = \nabla \cdot (\rho\, C\, \boldsymbol{v})_{\text{exact}} + \nabla \cdot (\rho\, C\, \boldsymbol{v})_{\text{error}} \tag{1}$$

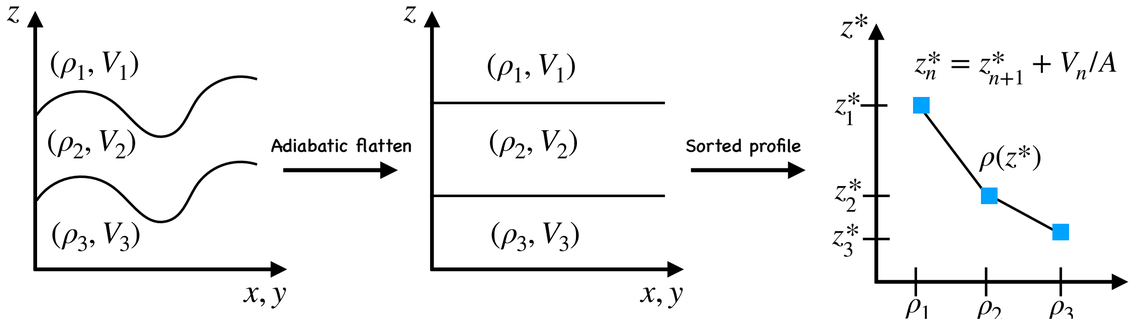
⋆ Errors in numerical advection can be interpreted as an extra SGS term

$$\frac{\partial(\rho\, C)}{\partial t} + \nabla \cdot (\rho\, C\, \boldsymbol{v})_{\text{exact}} = -\nabla \cdot \left[\boldsymbol{J} + (\rho\, C\, \boldsymbol{v})_{\text{error}}\right]. \tag{2}$$

- The error term is not physics, nor is it under our direct control (other than through numerical methods). If it is large (as is common), then it corrupts the physical integrity of the simulation.
- The error term is enhanced when strong variance accumulates at the grid scale, which can occur particularly in a mesoscale-turbulent simulation with strong gradients.
- Spurious mixing can be reduced by higher order numerics, but that is often insufficient when confronted with 100-1000s of eddy turnover times (as in climate simulations).
- Quasi-flow-following vertical coordinates (e.g., isopycnal or z-tilde) have proven effective at reducing spurious mixing.

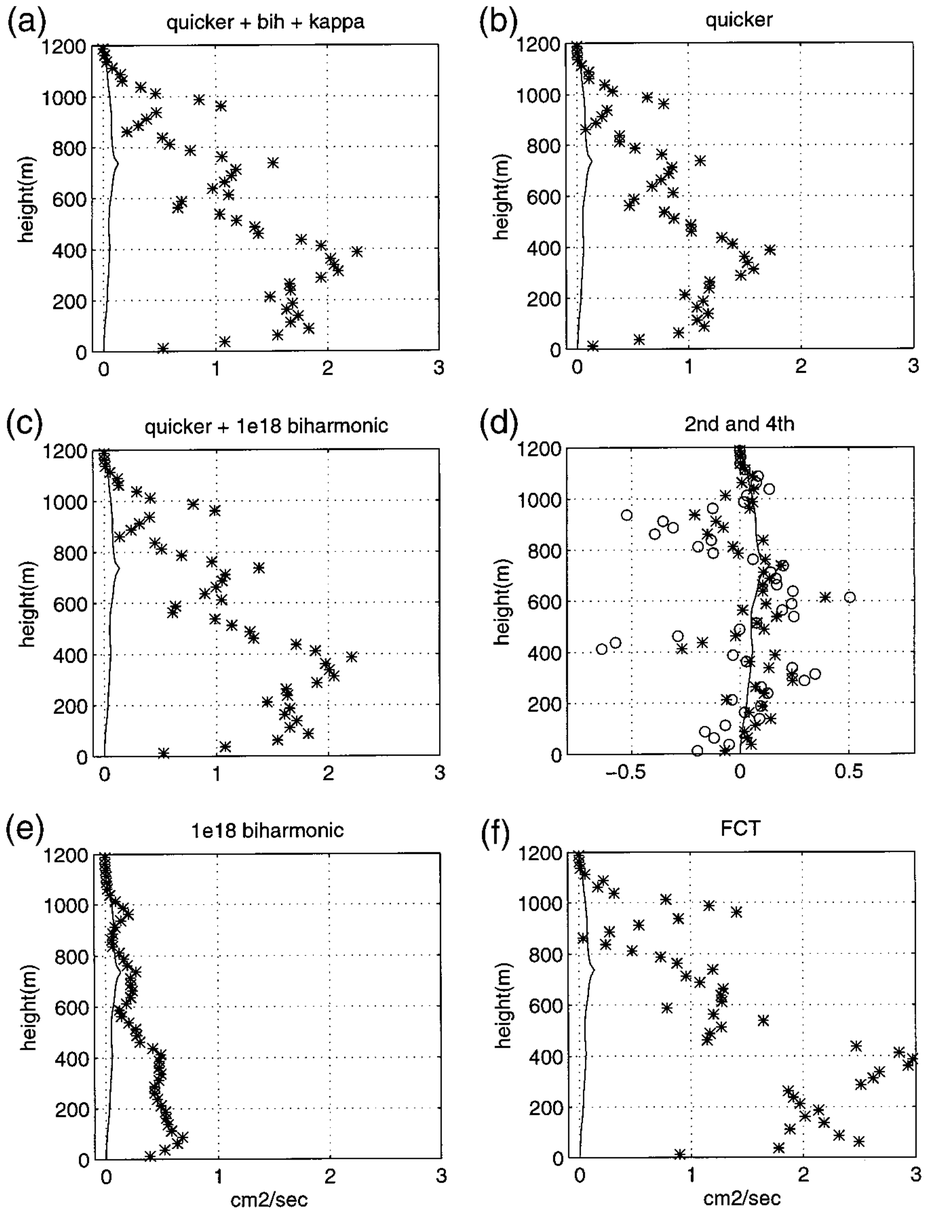
Sorting method to diagnose spurious numerical mixing (one of many methods)

Griffies, Pacanowski, Hallberg (Monthly Weather Review 2000)



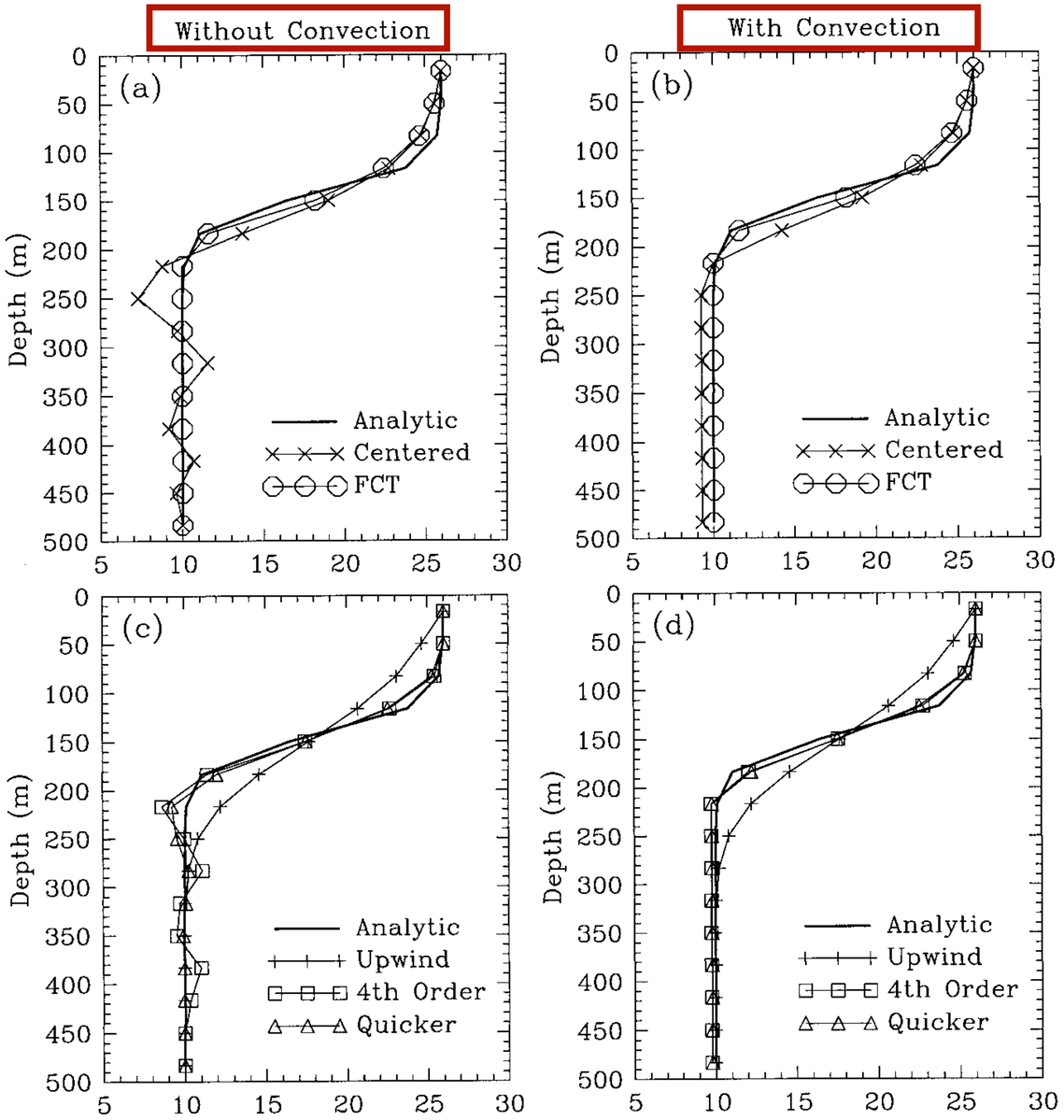
- ⋆ A method based on density sorting to produce a stable background profile, ρ_{back}, following Winters and D'Asaro (1995).
- ⋆ In an adiabatic simulation, evolution of the background state only arises from spurious numerical sources, which we interpret as an effective diffusivity, κ_{eff}

$$\frac{\partial \rho_{\text{back}}}{\partial t} = \frac{\partial}{\partial z^*} \left[\kappa_{eff} \frac{\partial \rho_{\text{back}}}{\partial z^*} \right]$$



- Levels of spurious mixing are generally far larger than physical mixing, even with "modern advection schemes".
- One trick is to add sufficient viscosity to ensure all flow scales are well resolved.
- Modelers are generally resistent to enhancing friction, since they like to see wiggles!
- But there is no free lunch, with (Ilicak et al 2012) arguing for order one grid Reynolds number to ensure flow scales are adequately resolved.

Example: thermocline spreading with advection/convection



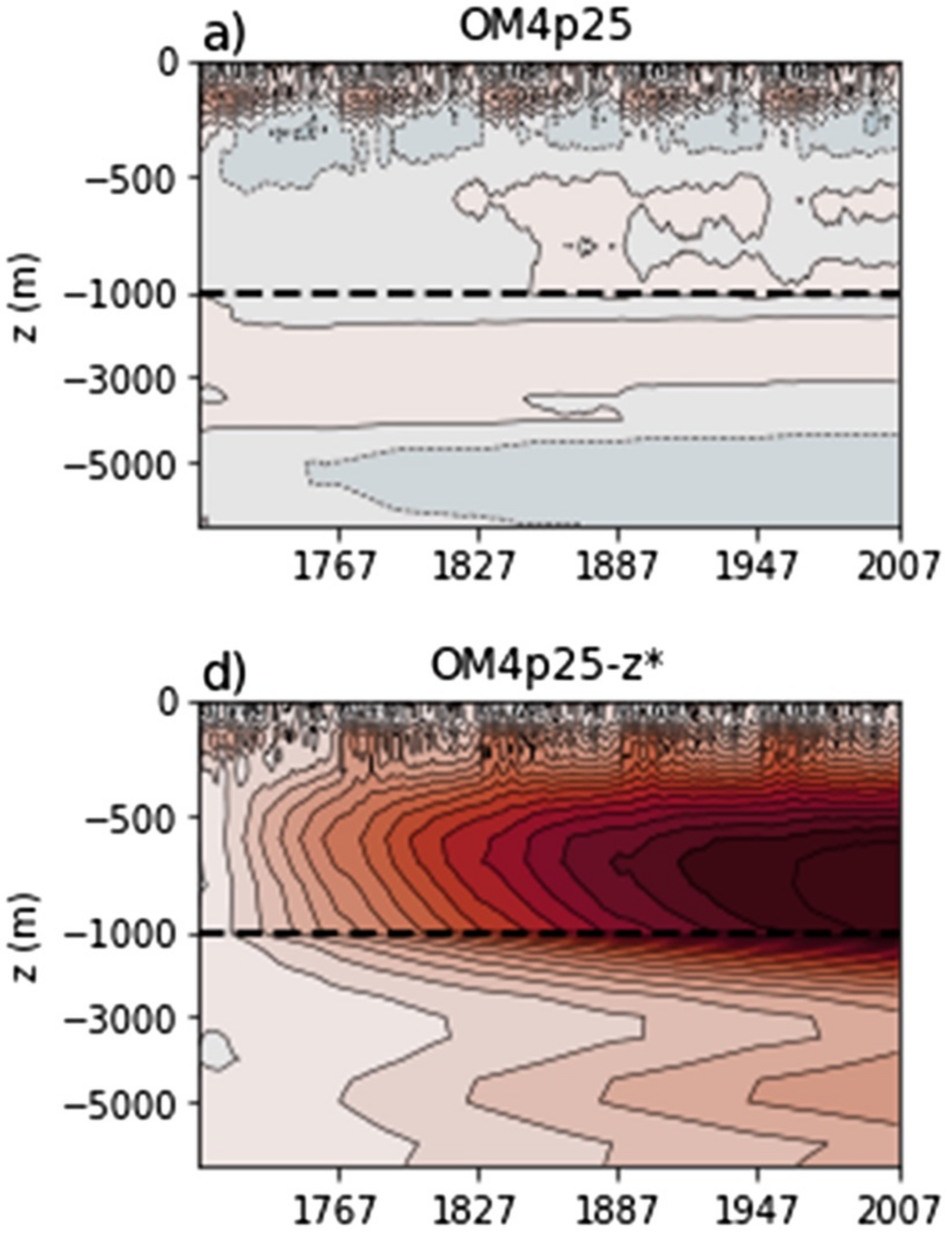
- Although non-diffusive, centered advection is very dispersive.
- Dispersion leads to over-shoots (unphysical extrema) and non-monotonic behavior
- For many years, adding diffusion/ friction was the preferred route to keeping solutions physically smooth and avoiding overshoots.
- Newer methods attack non-monotonicity in more sophisticated (local manners), such as flux-correction methods and WENO methods.

Comparing advection schemes on temperature in seasonally varying tropical thermocline. Adding convection smooths extrema but also diffuses the thermocline.

Griffies, Pacanowski, Hallberg (Monthly Weather Review 2000)

Importance of addressing the spurious mixing problem

- Depending on vertical coordinate choice, it can greatly reduce spurious mixing.
- Arbitrary vertical coordinates, including hybrid (we use z-rho hybrid).
- Here we ameliorated the spurious mixing by switching to isopycnal-like coordinates.
- Other approaches might work as well, but they are largely untested in the global climate context.



Adcroft et al 2019

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RESEARCH ARTICLE
10.1029/2019MS001726

Special Section:
Geophysical Fluid Dynamics Laboratory CMIP6 Models

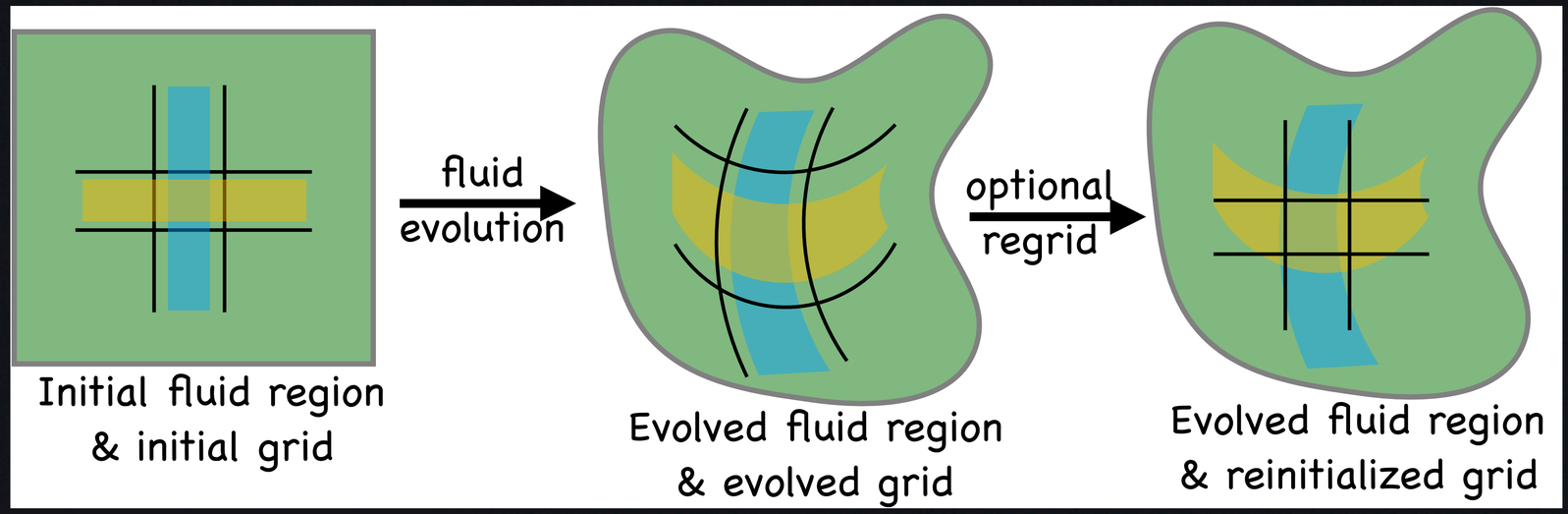
The GFDL Global Ocean and Sea Ice Model OM4.0: Model Description and Simulation Features

Alistair Adcroft[1,2], Whit Anderson[2], V. Balaji[1,2], Chris Blanton[2,3], Mitchell Bushuk[2], Carolina O. Dufour[4], John P. Dunne[2], Stephen M. Griffies[1,2], Robert Hallberg[1,2], Matthew J. Harrison[2], Isaac M. Held[1,2], Malte F. Jansen[5], Jasmin G. John[2], John P. Krasting[2], Amy R. Langenhorst[2], Sonya Legg[1,2], Zhi Liang[2], Colleen McHugh[2,3], Aparna Radhakrishnan[3], Brandon G. Reichl[2], Tony Rosati[2], Bonita L. Samuels[2], Andrew Shao[6], Ronald Stouffer[2,7], Michael Winton[2], Andrew T. Wittenberg[2], Baoqiang Xiang[2,8], Niki Zadeh[2,3], and Rong Zhang[1,2]

Key Points:
- Documentation is provided for a new generation of NOAA-GFDL CMIP6/OMIP ocean ice climate models

Lagrangian methods for numerical models

- Arbitrary Lagrangian-Eulerian (ALE) refers to any method that allows for moving grid cells.
- "Advection" refers to the transport of fluid across cell boundaries.
  - Step 1: Lagrangian step: grid moves with flow (no advection)
  - Step 2: Regrid/remap step to reorganize the grid back to a target (acts like advection)
- Ideally, step 2 has no impact on the fluid state, but in practice there are truncation errors that impact the representation (i.e., spurious mixing).



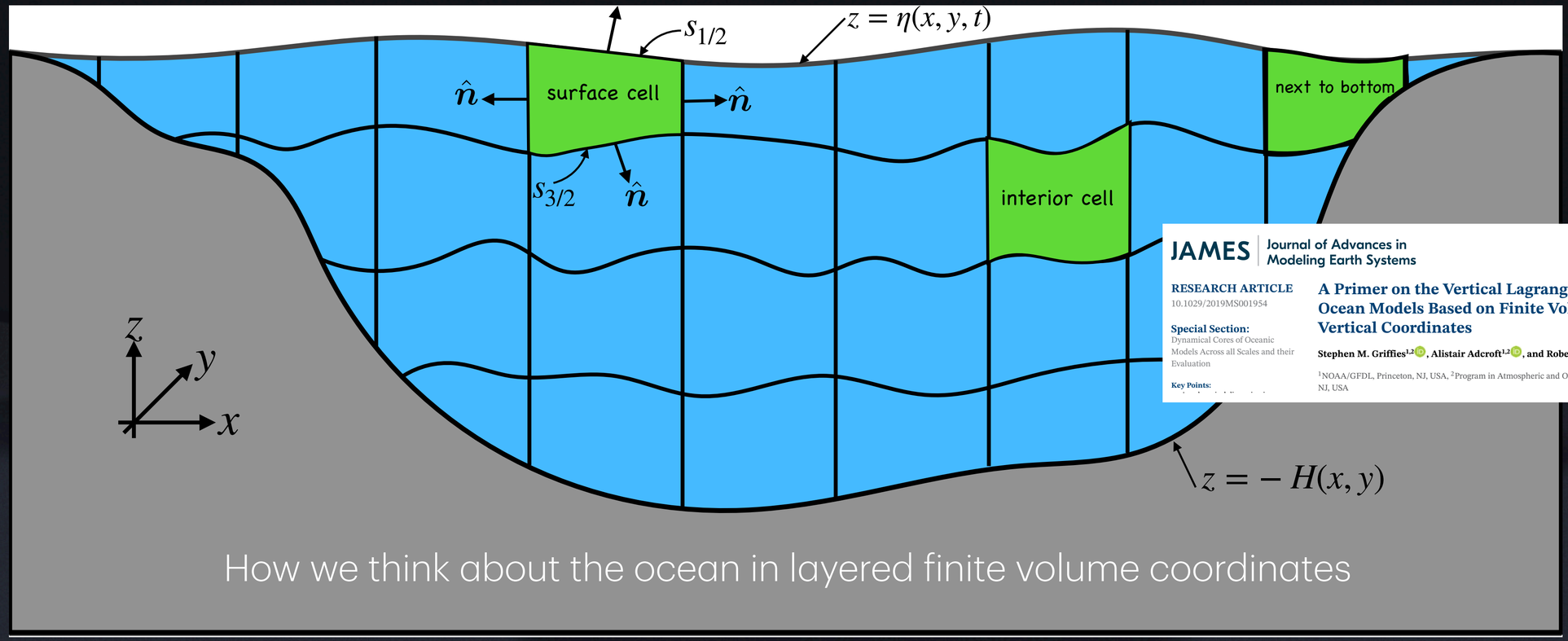
Griffies, Adcroft, Hallberg (2020) JAMES

The remap step is not necessary if the fluid regions remain coherent and only moderately distorted. But that is not the case for ocean flows, particulalry when they are turbulent and/or exhibit chaotic advection.

Many of the early ideas about Lagrangian methods started at Los Alamos in the 1970s as reviewed in Hirt, Amsden, Cook (1974)

Layered vertical coordinates to respect the stirring/mixing paradigm

- Modelers realize there are limitations to the tools we build, so the quest for the next "big thing" is relentless.
- Some ideas are old (e.g., use layers not levels), but can take decades to realize in a manner sufficient for realistic climate models.
- MOM6 has its roots in Hallberg's PhD thesis (from 1995), Bleck's HYCOM (2002), as wells MOM5 (Griffies) and MITgcm (Adcroft).



How we think about the ocean in layered finite volume coordinates

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RESEARCH ARTICLE
10.1029/2019MS001954

Special Section:
Dynamical Cores of Oceanic Models Across all Scales and their Evaluation

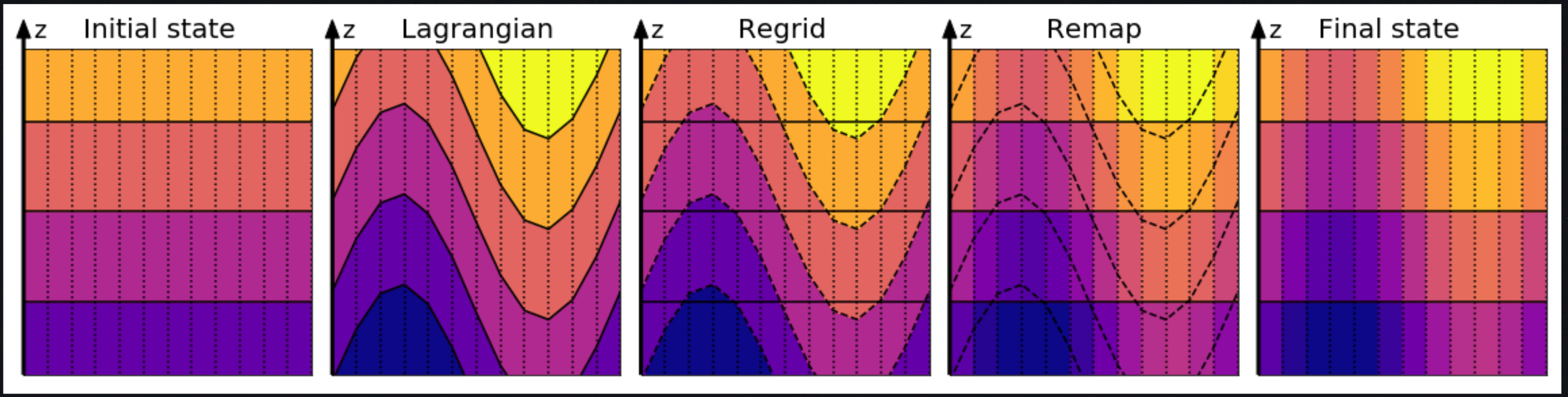
Key Points:

A Primer on the Vertical Lagrangian-Remap Method in Ocean Models Based on Finite Volume Generalized Vertical Coordinates

Stephen M. Griffies[1,2], Alistair Adcroft[1,2], and Robert W. Hallberg[1,2]

[1]NOAA/GFDL, Princeton, NJ, USA, [2]Program in Atmospheric and Oceanic Sciences, Princeton University, Princeton NJ, USA

Vertical Lagrangian remap method for ocean models



Griffies, Adcroft, Hallberg (2020) JAMES

- Vertical Lagrangian: zero vertical advection but nonzero horizontal advection.
- Vertical transport is realized by remapping. It is mathematically identical to vertical advection, but numerically it is very distinct.
- There is no CFL associated with vertical transport, so can have infinitesimally thin cells.
- Geophysical fluid models only do vertical Lagrangian; not 3d Lagrangian.
- It is strategically useful to design vertical coordinates the follow the flow. Method was pioneered for ocean modeling by Rainer Black (2002) in HyCOM.
- HyCOM development has today been merged with MOM6.

Spurious diapycnal mixing and vertical methods: summary points

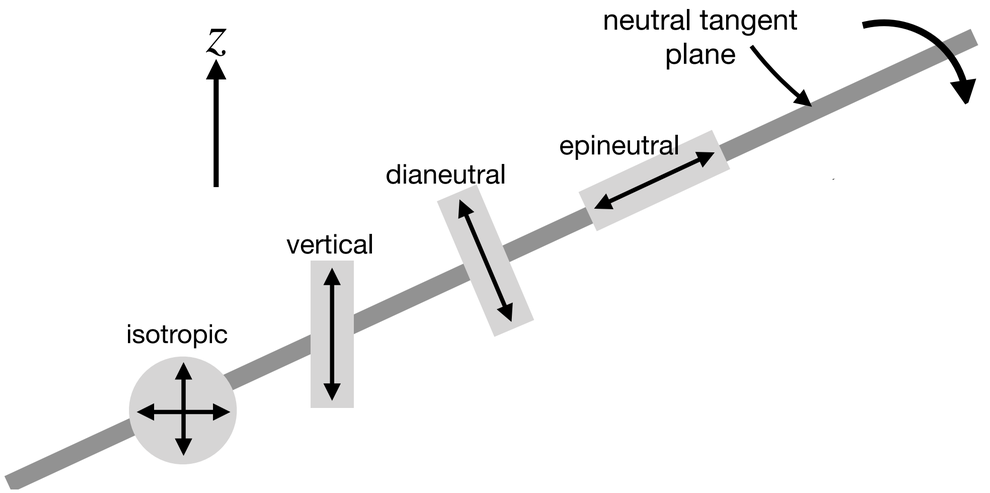
- Spurious diapycnal mixing happens, in varying quantities, within all ocean models.
- When integrated over 100s of years, the spurious mixing can accumulate over a huge part of the ocean to enhance interior mixing well beyond what is physically realistic.
- Mitigation requires smart vertical coordinates and accurate numerical transport operators.
- MOM6 uses hybrid z*-isopycnal with a vertical Lagrangian remap method.
- Ongoing issues:
    1. How to (locally) diagnose spurious diapycnal mixing. It is not trivial numerically. There are many methods (e.g., variance methods from Burchard and Klingbeil), but no one method satisfies all diagnostic needs.
    2. Constructing vertical coordinates: what is the best choice? We can do almost anything, but now we must decide what to use!!
    3. Enhancing numerical accuracy of advective transport operators: a timeless quest.

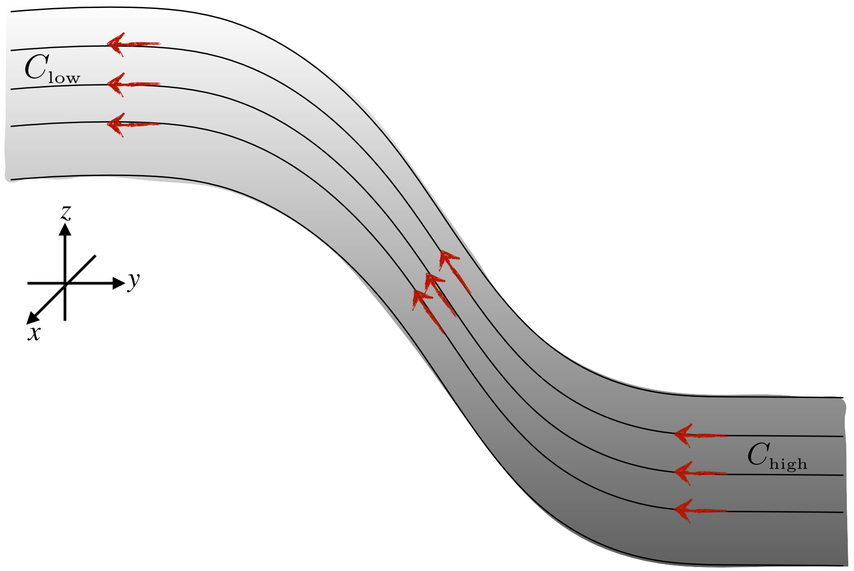
Part III: vertical heat transport via mesoscale eddies

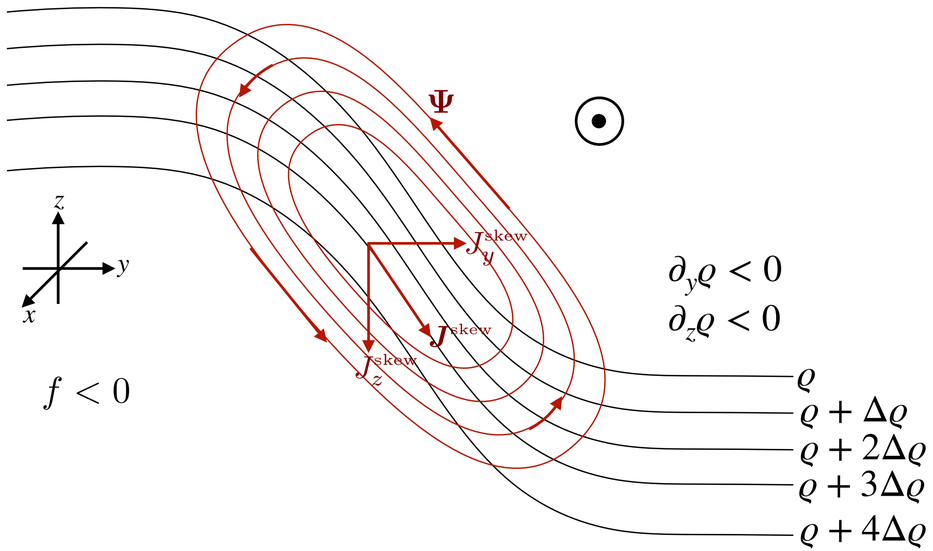
Role of ocean mesoscale eddies in vertical heat transport

Ocean eddies play a fundamental role in ocean heat uptake.

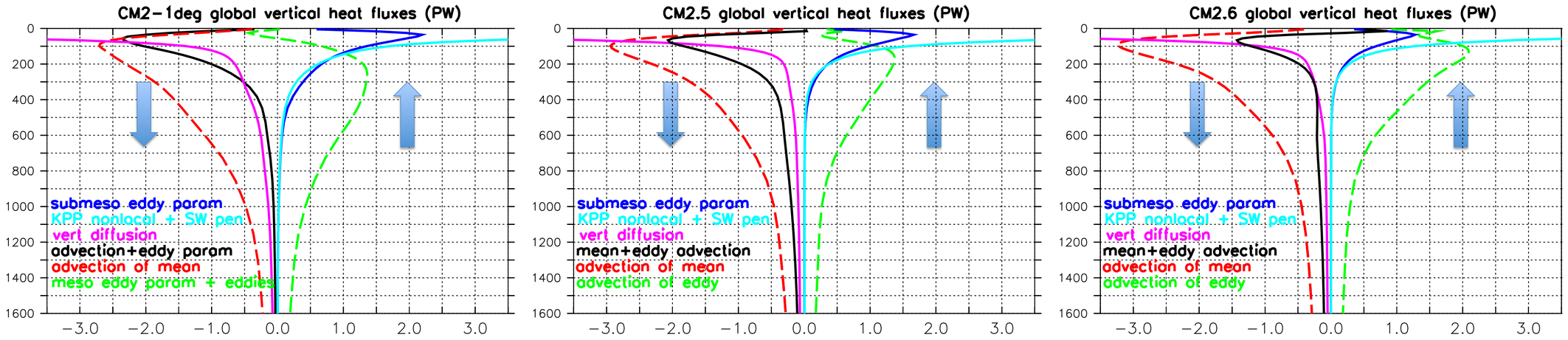
- Mesoscale eddies affect the vertical transfer of heat, thus playing a central role in thermal equilibration of climate simulations.
- Mesoscale eddies pump tracer variance to the grid scale, thus stressing the numerics and exacerbating the spurious mixing problem.







CM2.6/CM2.5/CM2-1deg model hierarchy illustrating the eddy effect

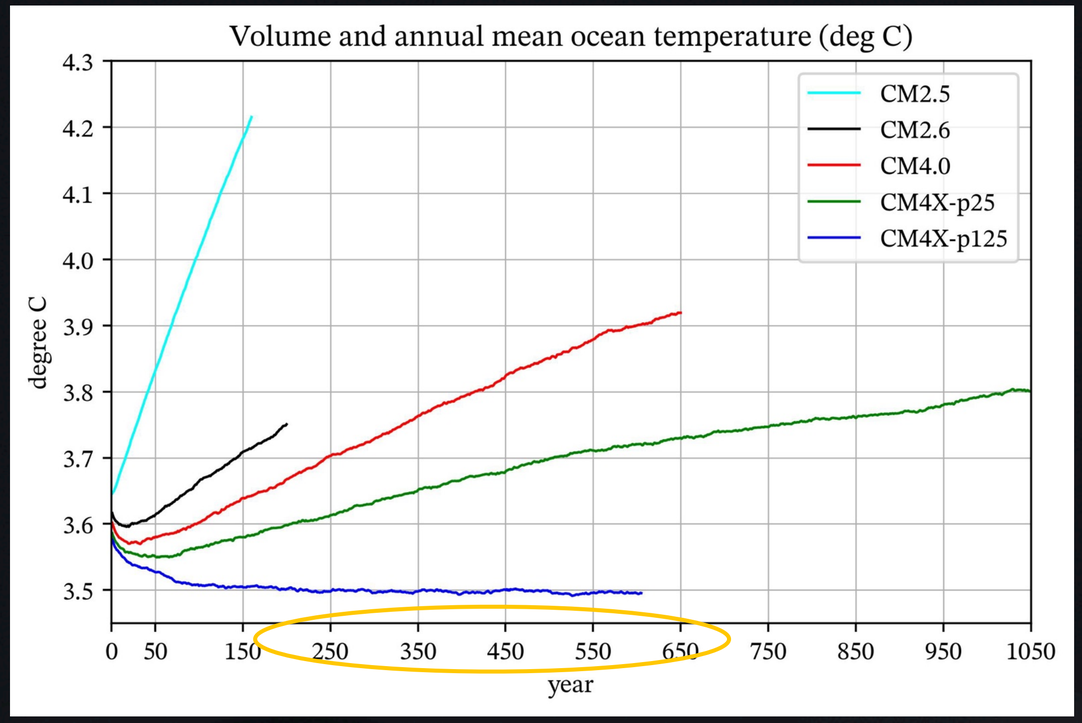


Griffies et al 2015

- Mean heat transport downward into ocean (mechanical forcing)
- Eddy heat transport upward, with CM2.6 roughly 50% larger than CM2.5 at intermediate depths.
- Eddy parameterization acts like eddies, but more like that in CM2.5 rather than CM2.6.
- Compensation is more complete for CM2.6; this model is more adiabatic.

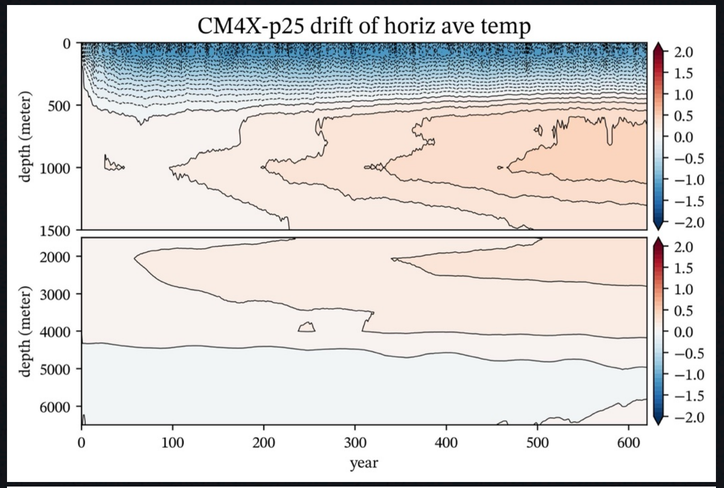
CM4X-hierarchy piControl thermal equilibration

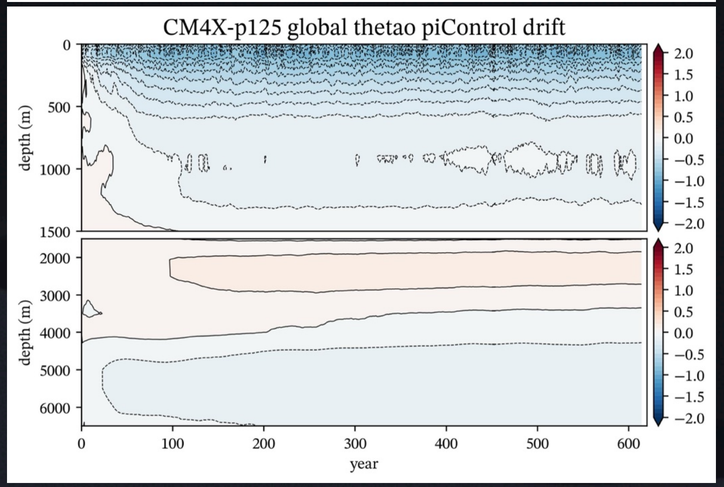
- CM4X-p125 thermally spins up within ~150 years to a steady state with -0.02W/m2 global imbalance
- CM4X-p25 takes ~1000 years, whereas CM2.5 and CM2.6 appear to need even longer (though they are far from equilibrated so it is tough to know).
- In the process of equilibrating, CM4X-p125 releases about 400ZJ of heat, which is comparable to the Zanna et al (2019) estimate of anthropogenic heat uptake since 1870.



Strong eddy effect + weak spurious mixing allows for CM4X-p125 to reach a thermal quasi-equilibrium in 150years! This is the first model to do that!

It is a big deal for climate modelers to have a realistic spin-up within 150 years!!





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RESEARCH ARTICLE
10.1029/2024MS004861

The GFDL-CM4X Climate Model Hierarchy, Part I: Model Description and Thermal Properties

Key Points:
- CM4X-p25 and CM4X-p125 are designed to study ocean and sea ice physics, focusing on effects from eddies and boundary currents
- CM4X-p125 reaches pre-industrial thermal equilibrium in 150 years whereas the coarser CM4X-p25 has yet to equilibrate after 1000 years

Stephen M. Griffies[1,2], Alistair Adcroft[2], Rebecca L. Beadling[3], Mitchell Bushuk[1], Chiung-Yin Chang[2], Henri F. Drake[4], Raphael Dussin[1], Robert W. Hallberg[1,2], William J. Hurlin[1], Hemant Khatri[5], John P. Krasting[1], Matthew Lobo[2], Graeme A. MacGilchrist[6], Brandon G. Reichl[1], Aakash Sane[2], Olga Sergienko[2], Maike Sonnewald[7], Jacob M. Steinberg[1], Jan-Erik Tesdal[2], Matthew Thomas[8], Katherine E. Turner[9], Marshall L. Ward[1], Michael Winton[1], Niki Zadeh[1], Laure Zanna[10], Rong Zhang[1,2], Wenda Zhang[2], and Ming Zhao[1]

Vertical heat transport via mesoscale eddies: summary points

- Weak vertical eddy heat transport happens when under parameterize and/or under represent mesoscale.
- Contributes to problems with ocean heat uptake, long-term model drift, etc.
- Mitigation requires smart eddy parameterizations/representations.

Part IV: a numerical method conserving potential vorticity in layered models

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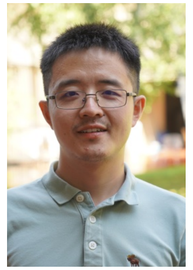
RESEARCH ARTICLE
10.1029/2025MS005376

Key Points:

- A WENO-based transport scheme for potential vorticity (PV) is implemented

A WENO Finite-Volume Scheme for the Evolution of Potential Vorticity in Isopycnal Ocean Models

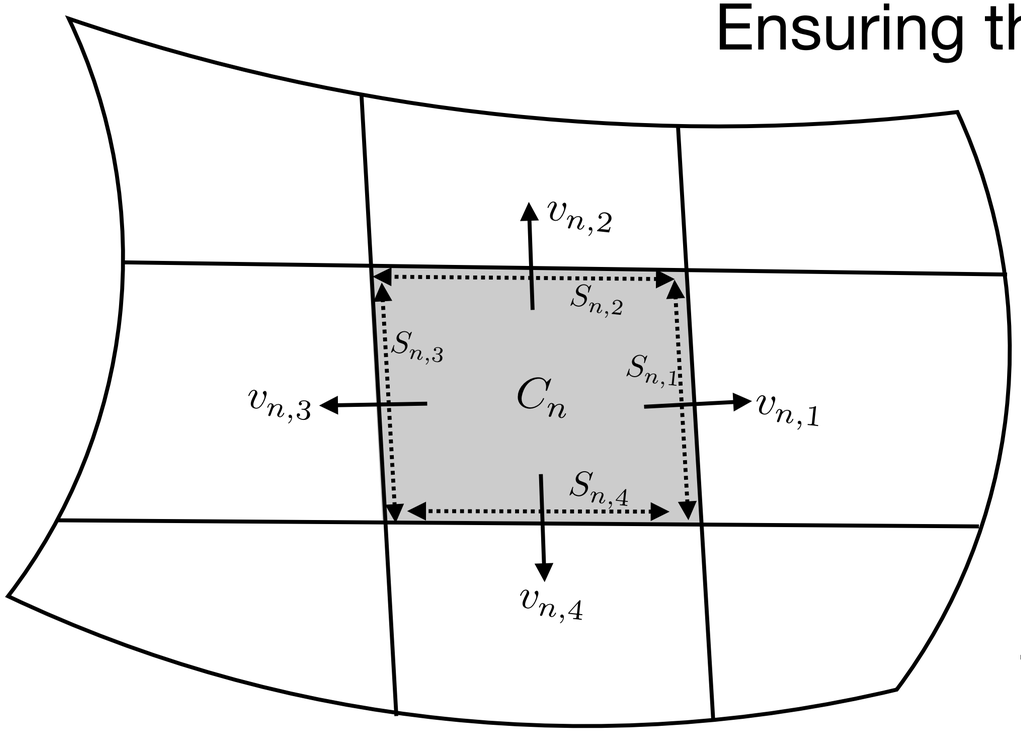
Wenda Zhang[1], Yi-Hung Kuo[2], Simone Silvestri[3,4], Alistair Adcroft[1], Robert Hallberg[1,5], and Stephen M. Griffies[1]



Slides courtesy of Wenda Zhang, formerly of Princeton University and now at the Laoshan (崂山实验室) Lab in Qingdao, China

Bryan (1966): centered differences (60 years!)

2-page paper!



Ensuring the first and second moments are conserved

$$\sum_j v_{n,j}\, S_{n,j} = 0 \Longrightarrow \frac{\mathrm{d}V_n}{\mathrm{d}t} = 0$$

$$(C\,\boldsymbol{v}\cdot\hat{\boldsymbol{n}})_{n,j} \approx \overline{C}_{n,j}\, v_{n,j} = \tfrac{1}{2}(C_n + C_j)\, v_{n,j}$$

$$\Longrightarrow \frac{\mathrm{d}}{\mathrm{d}t}\sum_n C_n V_n = 0 \qquad \Longrightarrow \frac{\mathrm{d}}{\mathrm{d}t}\sum_n C_n^2 V_n = 0$$

Bryan, Manabe, Smagorinsky 1969

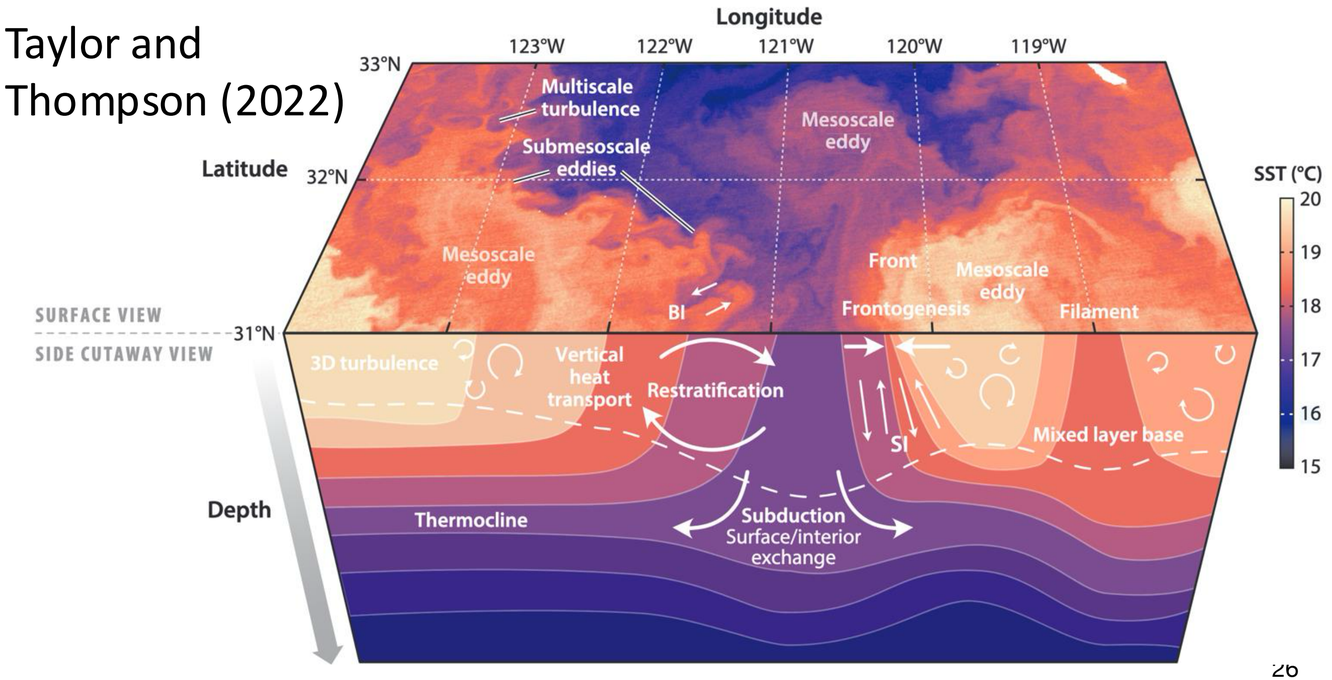
These properties formed the foundation for many numerical ocean models (e.g., MOM, NEMO) from the 1960s until the 2000s (nearly 50 years!).

- Removed the Phillips (1959) nonlinear instability by ensuring kinetic energy preserved by nonlinear advection.
- Simple numerics that are applicable to other operators.
- Vorticity methods later pursued by Arakawa/Lamb (1981) and collaborators.

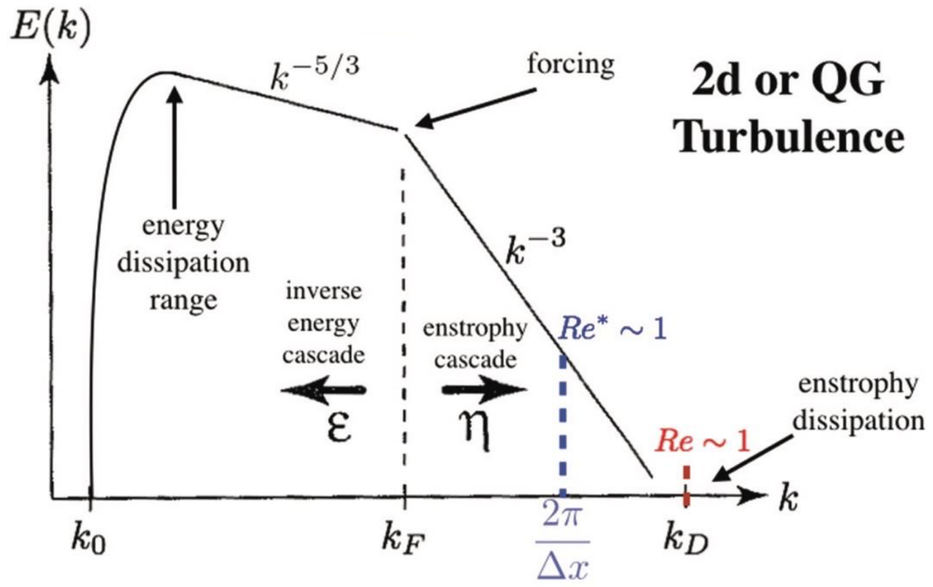
Eddies and cascades

- Ocean circulations involve multiple scales of motions, which interact with each other and transfer energy across scales
- At scales where geostrophic balance holds, energy tends to cascade upscale while potential enstrophy cascades downscale
- Ocean climate models can partially resolve mesoscale eddies, requiring subgrid closures to account for the cascade from/toward unresolved eddies

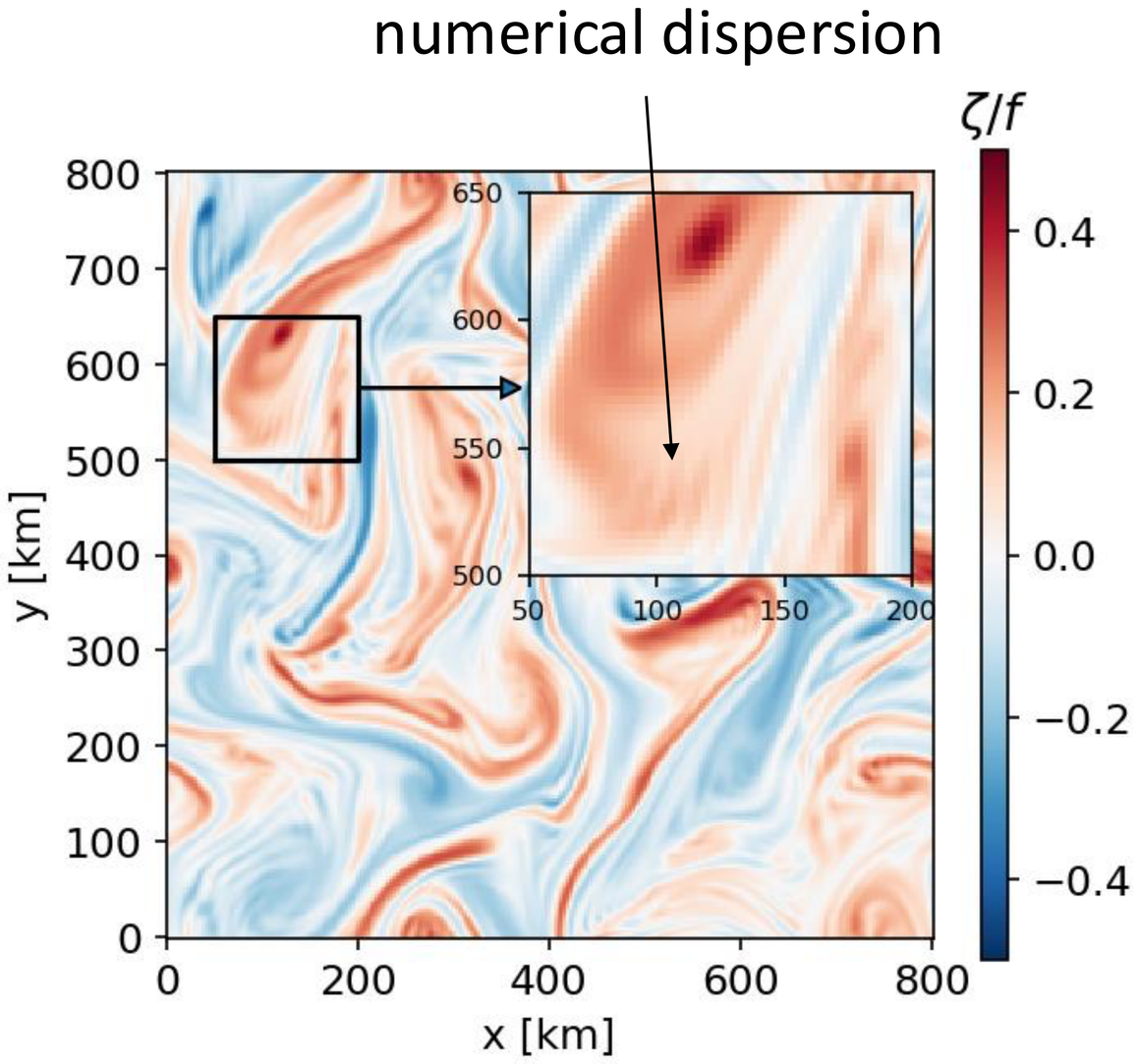
Taylor and Thompson (2022)



Fox-Kemper et al. (2014)



Numerical schemes for transport



- Ocean models often use **second-order momentum advection schemes** that conserve energy/enstrophy
- Such schemes contain strong **numerical dispersive errors** that need to be suppressed by a **large diffusivity/viscosity**

- Recently, many ocean models (e.g., ROMS, NEMO5.0, Oceananigans) have chosen to use **high-order upwind momentum advection schemes** that contain an **implicit numerical diffusion** to damp dispersive errors and grid-scale noises

- However, **momentum flux is not always downgradient** → numerical diffusion is not always a valid representation

Focusing on potential vorticity

- We now consider potential vorticity, with a focus on layered-isopycnal models.
- Potential vorticity requires one to pay attention to vorticity as well as boundaries.
- We do not time step potential vorticity in primitive equation models. Instead, its time evolution is diagnosed.
- What we produce is a discrete momentum equation that implies a conserved diagnosed potential vorticity equation in the inviscid limit.
- Initial tests suggest that the new scheme does play a role in reducing some errors related to interactions with topography.

PV dynamics in isopycnal ocean models

- Dynamical equations on isopycnal coordinate:

$$\partial_t u - vhq + \partial_x(M + \mathrm{K}) = \chi,$$
$$\partial_t v + uhq + \partial_y(M + \mathrm{K}) = \Upsilon,$$
$$\partial_t h + \partial_x(uh) + \partial_y(vh) = 0,$$

where $q = \frac{f+\zeta}{h}$ is the PV, $\mathrm{K} = 1/2(u^2 + v^2)$, and χ, Υ include the forcing and dissipation

- Taking curl and assuming $\hat{\chi} = \hat{\Upsilon} = 0$ yields the PV equation

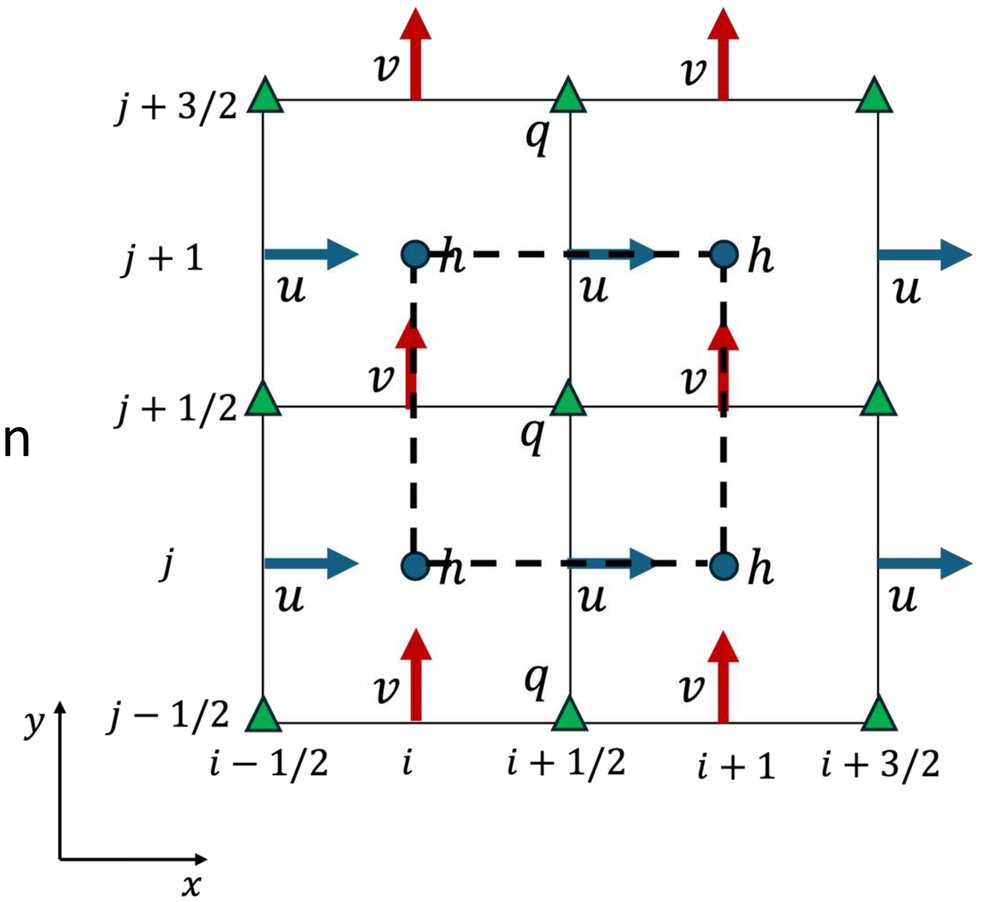
$$\frac{\partial hq}{\partial t} = -\nabla \cdot (\mathbf{u}hq)$$

- Discretization for the PV flux at *u* and *v* grids,

$$(vhq)_u = (vh)_u q_u + E_u$$
$$(uhq)_v = (uh)_v q_v + E_v$$

where E_u and E_v are truncation errors due to the finite-volume reconstruction

Arakawa C-grid layout



WENO discretization of PV fluxes in MOM6

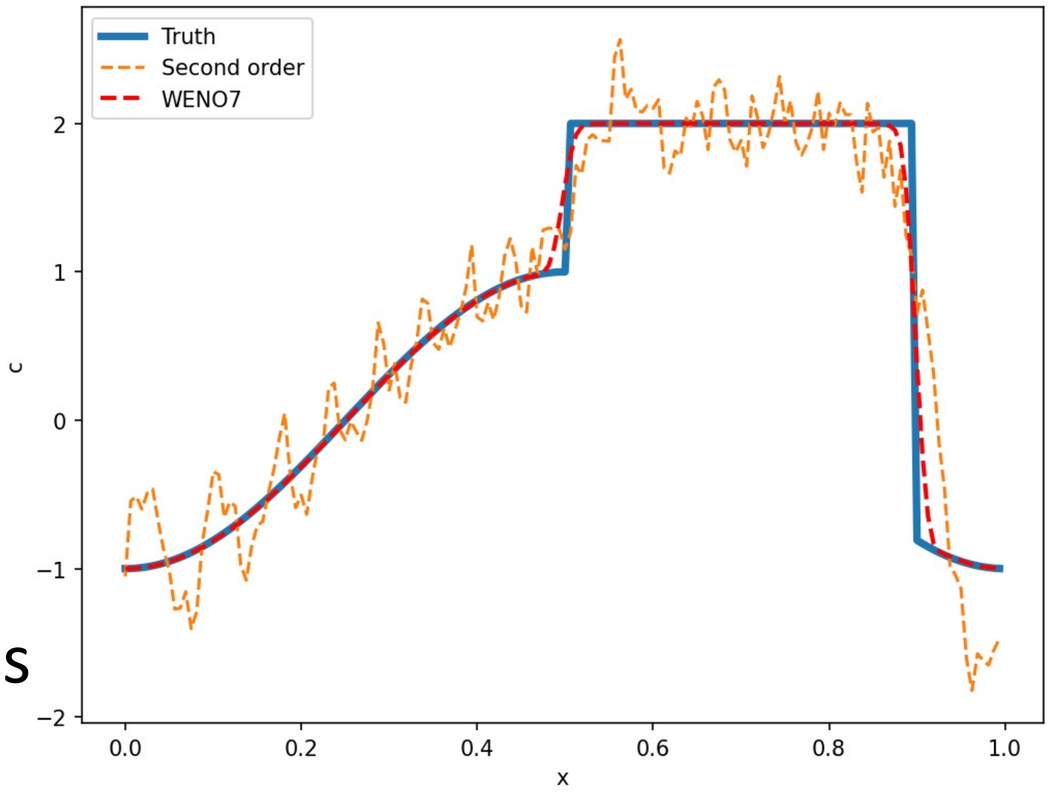
- Advection scheme for PV in MOM6:

➢ Original scheme: 2rd-order energy/enstrophy (**EEN**) conserving -> **dispersive errors**

➢ **We develop a new scheme**: 7th-order weighted essentially non-oscillatory (**WENO**) -> **diffusive errors**

➢ WENO minimizes the spurious oscillations and contains a very weak numerical diffusion

∂c/∂t+ ∂(uc)/∂x=0



Goal of this study

- Design a discretization scheme for **potential vorticity (PV) flux** that contains a weak **numerical PV diffusion** to represent the subgrid mixing

- The scheme is expected to **dissipate potential enstrophy** at grid scales while **maintaining an inverse energy cascade**

Wind driven tests

Setup:

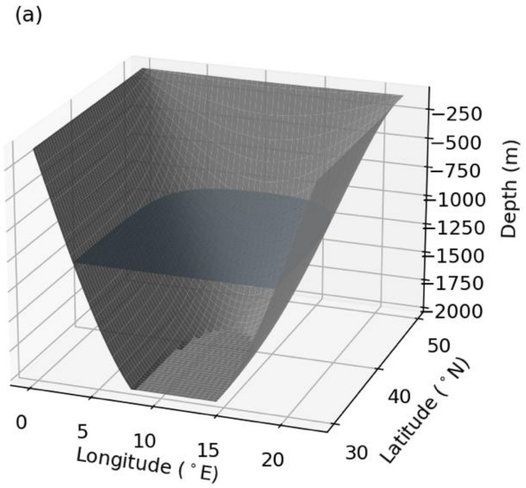
- Two isopycnal layers; Forced by constant zonal wind stress
- Sloping topography on lateral boundaries

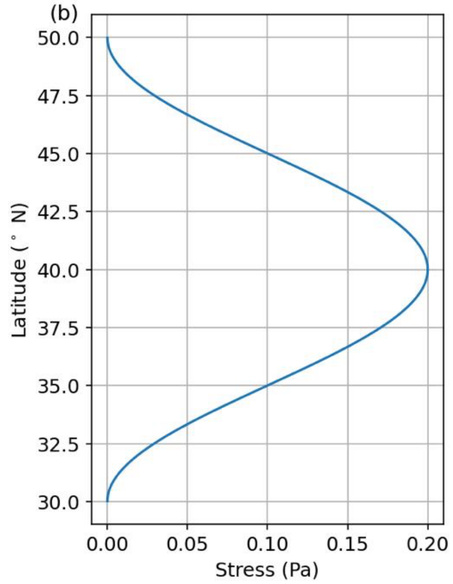
Experiments:

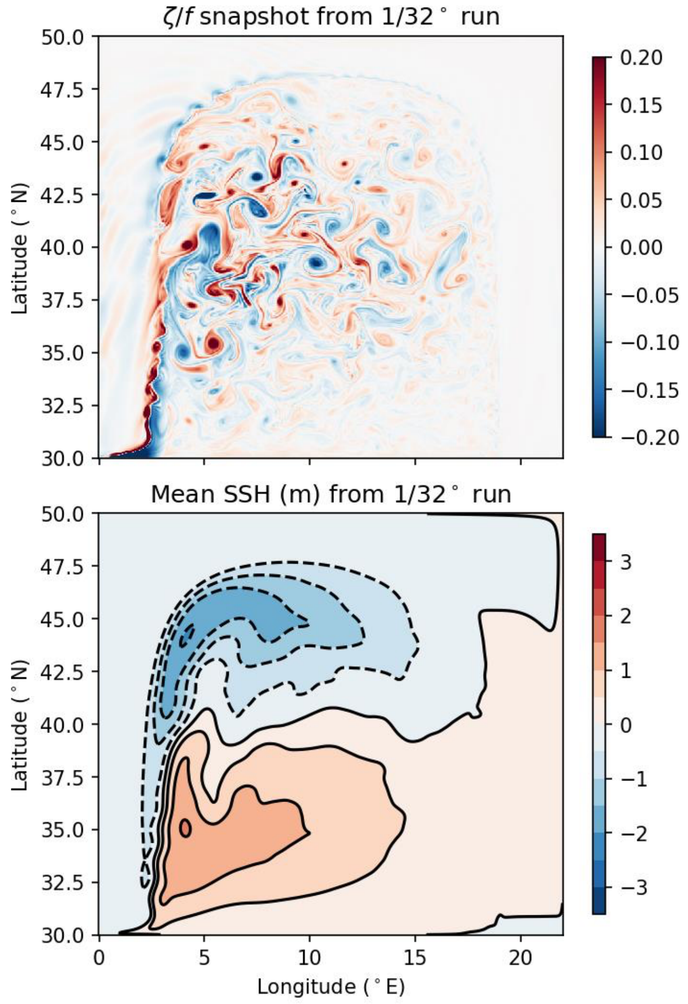
- **Control**: Second-order EEN scheme + biharmonic Smagorinsky viscosity
- **WENO7V**: Seventh-order WENO scheme without lateral viscosity

Computational cost:

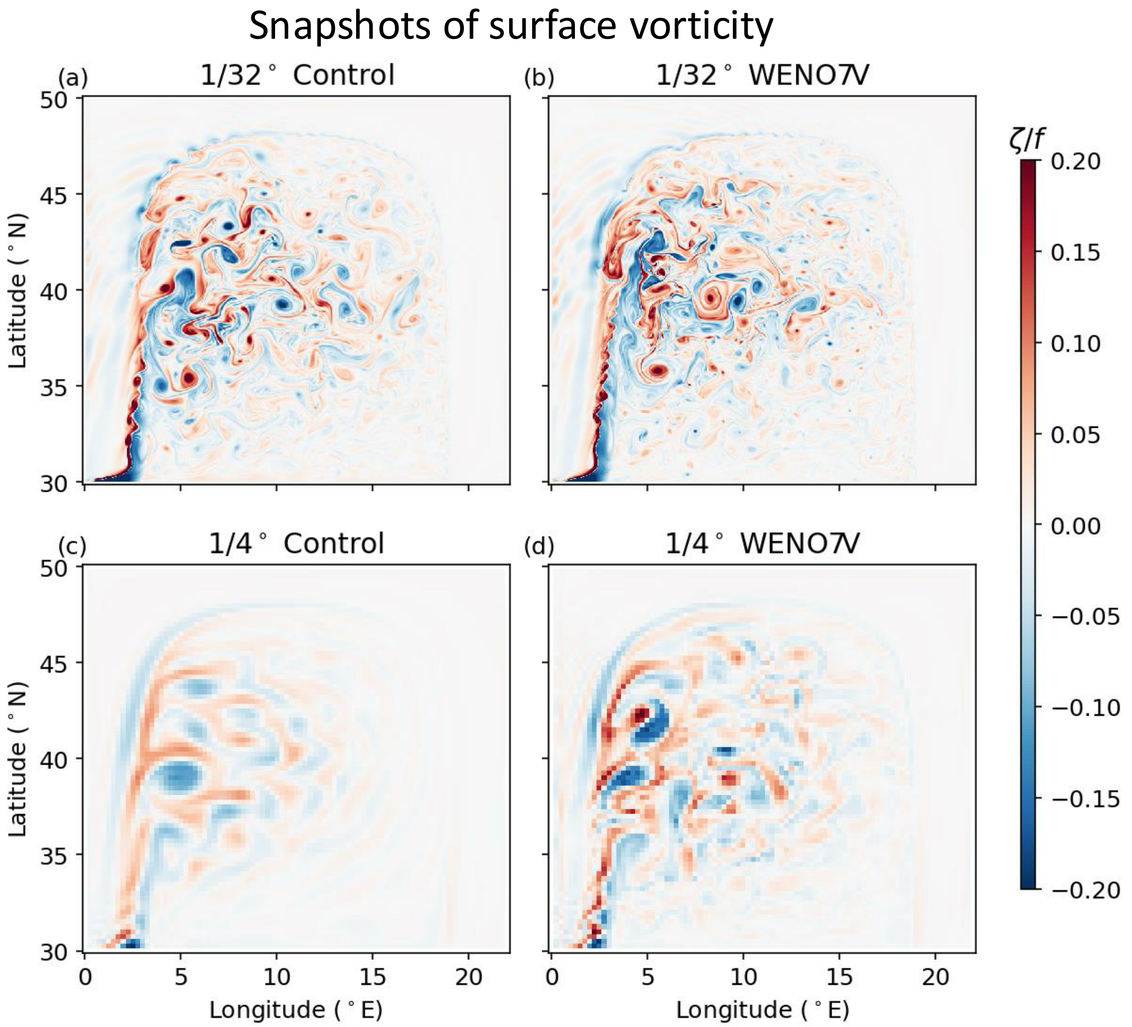
- The seventh-order WENO scheme incurs an additional **10-20%** cost to the entire model



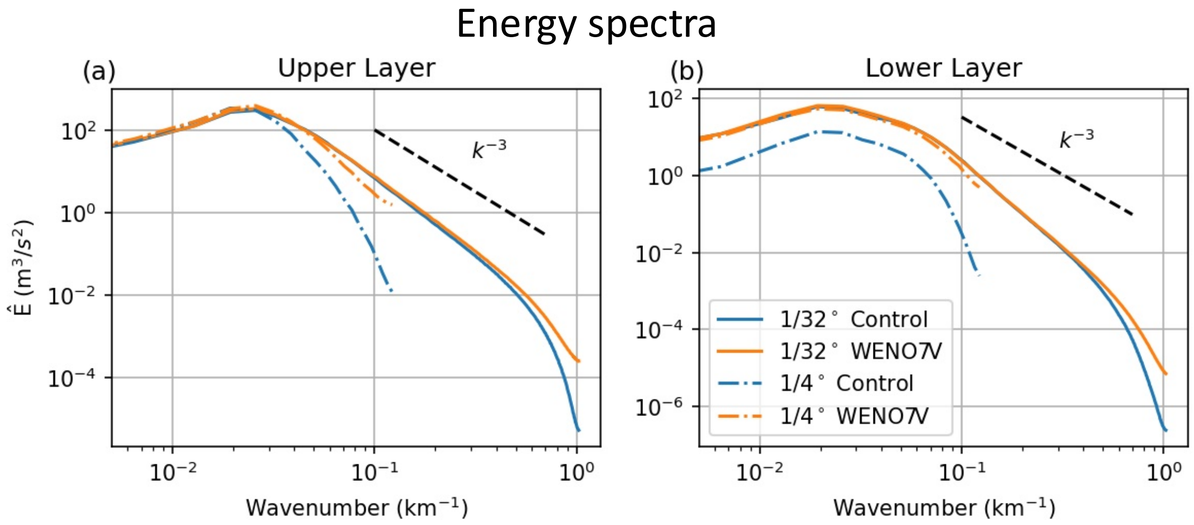




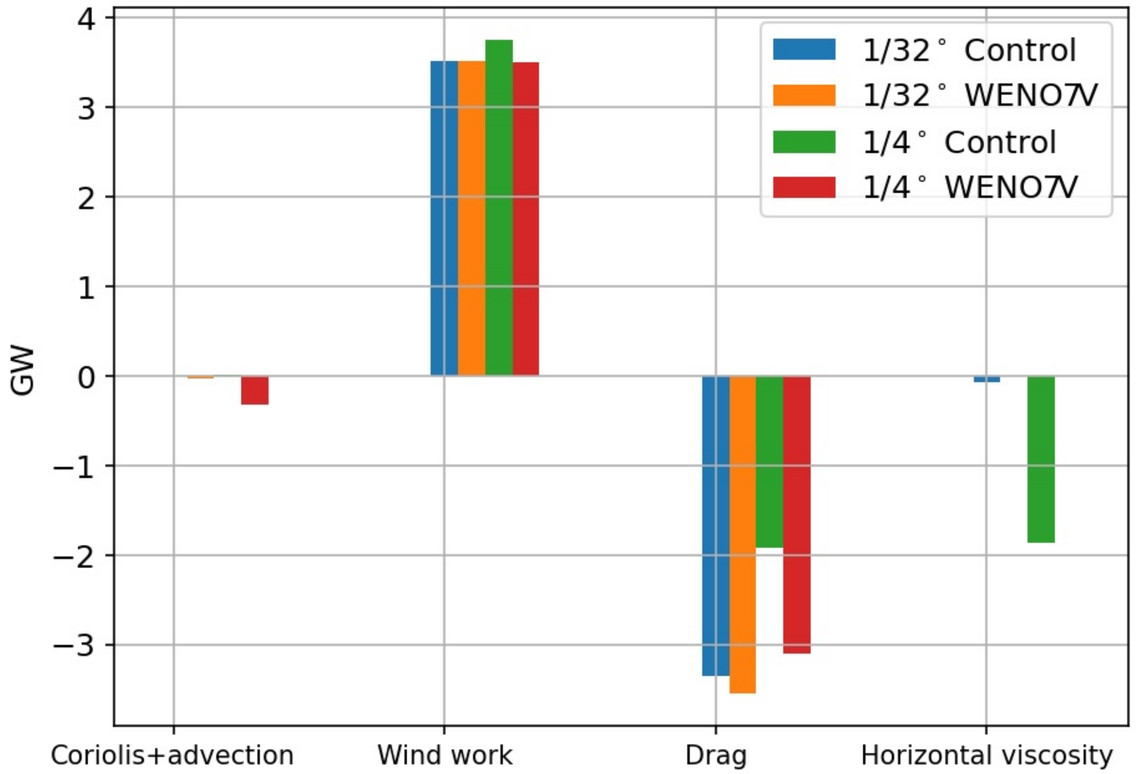
Eddy field



- At $1/32°$ resolution, WENO scheme does not make a big difference
- At $1/4°$ resolution, the WENO experiment exhibits **more energetic and finer-scale eddy structures**, closer to the $1/32°$ results

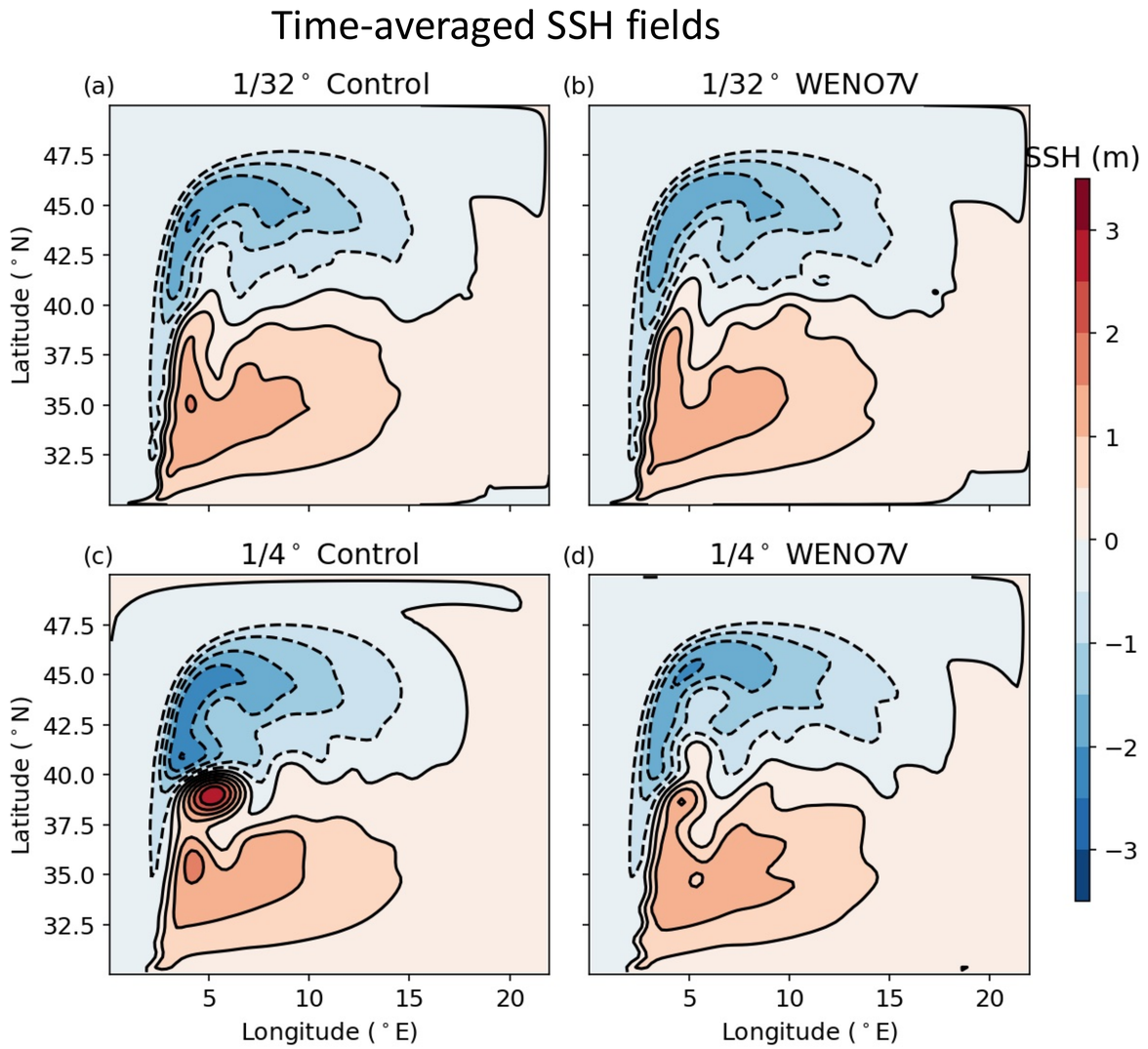


Kinetic energy budget



- The wind work is primarily dissipated by bottom drag at $1/32^\circ$ experiments
- At $1/4^\circ$ experiments, the dissipation by lateral diffusion is greatly reduced by the WENO scheme

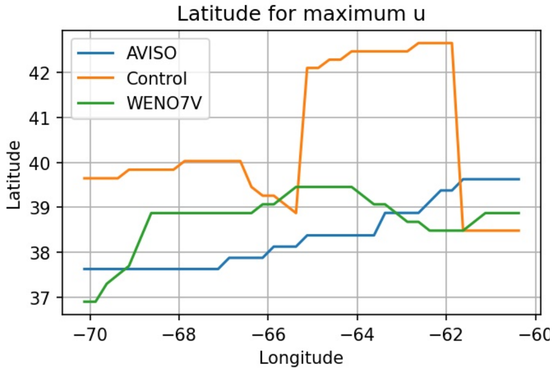
Large scale gyre circulation



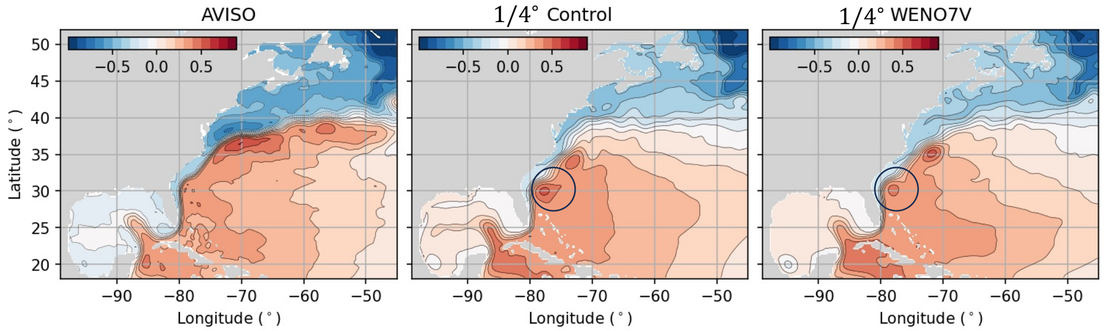
- Gyre circulations in the $1/4°$ WENO experiment are closer to $1/32°$ results
- The spurious standing eddy in the $1/4°$ control run is eliminated with the WENO scheme

Early tests in a global hybrid coordinate model

- The spurious standing eddy near Charleston Bump is reduced in the WENO experiment
- Gulf Stream separation point is more southward

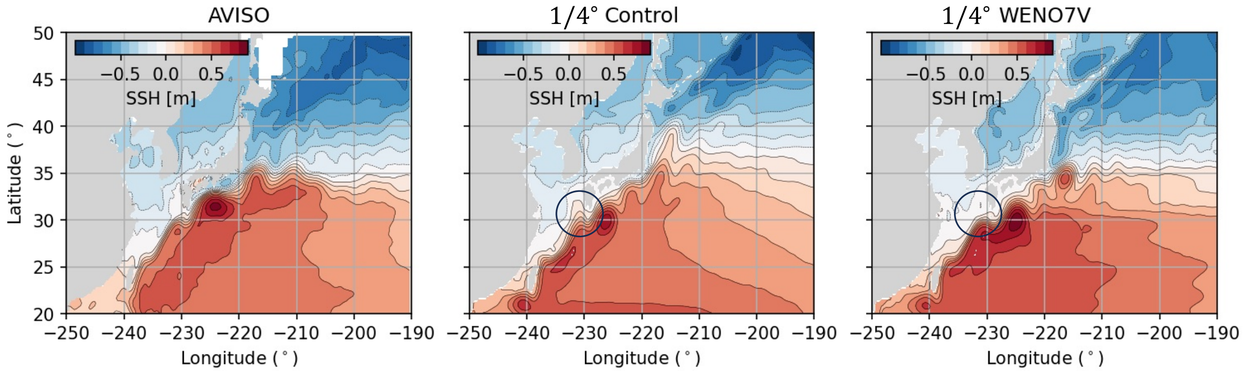


Annual-mean SSH field over 2003-2022 in Gulf Stream region



- The spurious meander at 30° N (Tokara Strait) is eliminated in the WENO experiment
- Kuroshio extension is more southward and Oyashio intrudes further southward in the WENO experiment, closer to observations

Annual-mean SSH field over 2003-2022 in Kuroshio region



PV advection summary points

- A **WENO PV advection scheme** is designed for the PV dynamics in isopycnal ocean models
- It represents an **implicit closure for subgrid PV mixing**
- **Eddy energetics** is improved with the new scheme, leading to enhanced kinetic energy and more realistic gyre circulations
- In preliminary tests in global ocean simulation, the scheme reduces biases in the **western boundary current pathways**
- Impact on other large-scale circulation features still need to be examined in the global ocean simulation

Closing comments and aspirations

Closing comments

- Numerical methods should aim to respect conservation properties of the continuum.
- Mundane properties such as tracer conservation are the baseline for local (grid cell) and global conservation. Not all modelers check for such properties, but they should!
- We also must give attention to the conservation of properties within isopycnal layers, in order to avoid unphysically large spurious mixing and the attendant problems with spurious heat and carbon transport and drift.
- Properties of non-prognosed properties, such as vorticity and potential vorticity, are hypothesized to be key to the integrity of boundary currents, flow-topography interactions, eddy dynamics. See Styles et al (2022) for problems with vorticity in C-grid, z-models such as NEMO and Oceananigans!
- The ongoing quest for numerical models that preserve higher moments (e.g., Casimirs of potential vorticity) must also remember the mundane properties. In particular, they must keep spurious mixing to a minimum! Otherwise, the models will produce beautiful movies and spectra, yet be of little use for climate simulations that must run for 1000s of eddy turnover times and incorporate complex boundaries.
- Special acknowledgement of the work from Peter Korn at MPI, who is pushing the resolution in global models to the non-hydrostatic regime. All of this talk remains relevant even at the O(100m) grid scales he is targeting!

Some aspirations for next few years of my career

- **Enhanced role of variational methods for numerical methods and analysis**
  - Hamilton's principle is making more inroads into the community. I wish to ride the tide and contribute as I can.
  - Apply to wave-mean flow interactions (particularly at meso/submeso/gravity wave scales) and to numerical methods.
  - First, I need to overcome an intellectual gap between canonical (Lagrangian) and non-canonical (Eulerian) methods.
  - Contact with the Imperial group (and Ollie Street in Paris) will greatly help in these aspirations!
- **Enhanced role for understanding vorticity/Ertel potential vorticity in regional + global models**
  - Diagnostic methods for PV (see Morel et al 2019 and Morel et al 2023)
  - How well can we develop an online budget?
  - Numerical methods to respect vorticity and render a sensible diagnostic field (see Styles et al. 2022 for a serious warning)
  - Formulate ocean circulation in PV-Theta space (inspired by Hoskins 1991 work in the atmosphere).
  - Move beyond planetary geostrophic PV (pioneered by Bryan 1987 in coarse models).
- **Cross-scale physical processes for global ocean circulation**
  - Diagnostic methods based on coarse-graining are now becoming common (Storer et al. 2022, Buzicotti et al. 2023, Aluie et al 2027).
  - Can they be useful as the basis for developing parameterization tools to support understanding the mesoscale-submesoscale-gravity wave continuum?
- **Bulk of the work is planned in collaboration (led by!) two postdocs starting early 2027**
  - Oliver Street from Imperial College
  - Jamie Hilditch from Stanford University
- Also, I wish to finish a (never-ending) book project on geophysical fluid mechanics!

Thanks for your attention!

Views from the Scotia Sea during April/May 2017 from the RRS JC Ross

Even theorists can go to sea, and they should!